

## Homogeneous Catalysis

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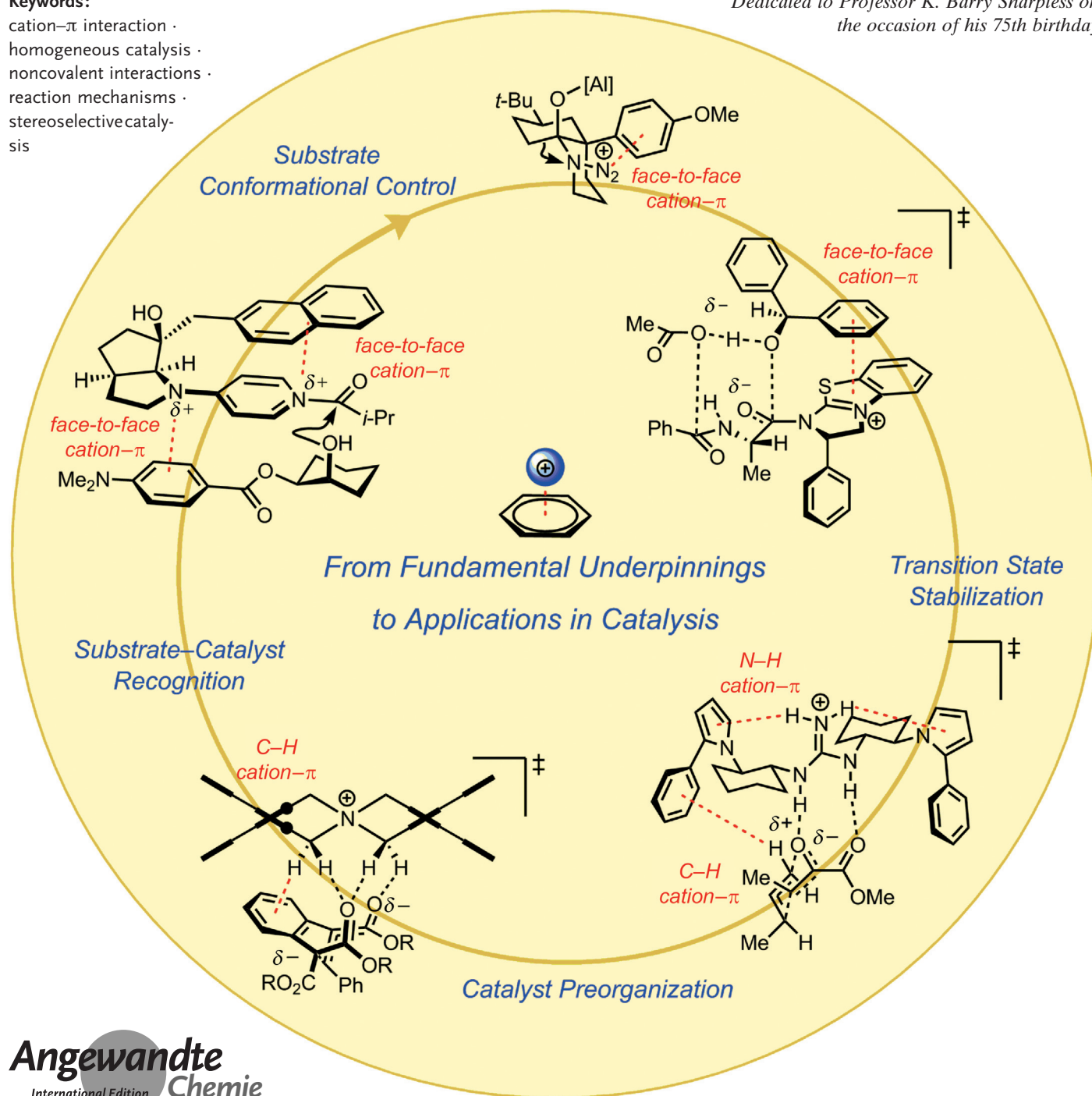
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The Cation- $\pi$  Interaction in Small-Molecule CatalysisC. Rose Kennedy<sup>+</sup>, Song Lin<sup>+</sup>, and Eric N. Jacobsen<sup>\*</sup>

## Keywords:

cation- $\pi$  interaction ·  
homogeneous catalysis ·  
noncovalent interactions ·  
reaction mechanisms ·  
stereoselective cataly-  
sis

Dedicated to Professor K. Barry Sharpless on  
the occasion of his 75th birthday



**C**atalysis by small molecules ( $\leq 1000$  Da,  $10^{-9}$  m) that are capable of binding and activating substrates through attractive, noncovalent interactions has emerged as an important approach in organic and organometallic chemistry. While the canonical noncovalent interactions, including hydrogen bonding, ion pairing, and  $\pi$  stacking, have become mainstays of catalyst design, the cation– $\pi$  interaction has been comparatively underutilized in this context since its discovery in the 1980s. However, like a hydrogen bond, the cation– $\pi$  interaction exhibits a typical binding affinity of several  $\text{kcal mol}^{-1}$  with substantial directionality. These properties render it attractive as a design element for the development of small-molecule catalysts, and in recent years, the catalysis community has begun to take advantage of these features, drawing inspiration from pioneering research in molecular recognition and structural biology. This Review surveys the burgeoning application of the cation– $\pi$  interaction in catalysis.

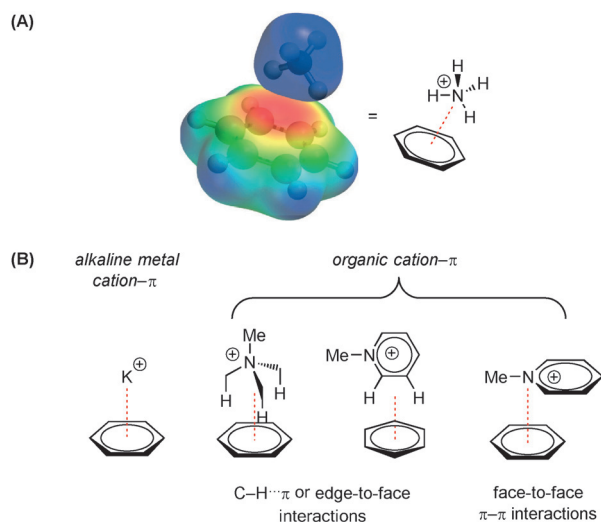
## 1. Introduction

The cation– $\pi$  interaction is a noncovalent attractive force between an electron-rich  $\pi$ -system (e.g., benzene, ethylene, acetylene) and a proximal cation (e.g.,  $\text{Na}^+$ ,  $\text{NBu}_4^+$ ; Figure 1).<sup>[1]</sup> Since its discovery in the 1980s,<sup>[2]</sup> experimental and computational methods have been applied to quantify and further understand the basis for and the magnitude of this interaction in both gas and condensed phases.<sup>[3]</sup> In this vein, the early discovery that a molecule of benzene could out-compete water to bind  $\text{K}^+$  in the gas phase—the association enthalpy of  $\text{K}^+$  and benzene being  $19 \text{ kcal mol}^{-1}$  whereas that

of  $\text{K}^+$  and water is  $18 \text{ kcal mol}^{-1}$ —was particularly noteworthy.<sup>[2a,4]</sup>

The thermodynamic features of this attractive interaction have also been studied in solution.<sup>[5]</sup> Although the strengths of cation– $\pi$  interactions are attenuated in solution relative to the gas phase, they are of comparable magnitude to other critically important, strong noncovalent interactions, including hydrogen bonds and ion pairs.<sup>[6]</sup> The significant binding energy is primarily attributed to the electrostatic attraction of a cation to an electron-rich  $\pi$ -face,<sup>[7]</sup> while dispersion and charge-transfer terms have been proposed to play contributing roles.<sup>[8]</sup> Furthermore, the cation– $\pi$  interaction exhibits a high degree of directionality, wherein the cation has a pronounced energetic preference for resting directly over the centroid of the  $\pi$ -system, thereby maximizing charge and orbital interactions.<sup>[5f,9]</sup>

The impact of these seminal studies on the field of structural biology was rapid and significant. A host of examples of the cation– $\pi$  interaction in biological macromolecules, such as protein-based receptors and enzymes, have been documented.<sup>[10]</sup> This ongoing work provides much of the framework for understanding how cation– $\pi$  interactions can serve structural and functional roles as these interactions are typically found to be responsible for either a) controlling protein conformation to generate a ligand- or substrate-binding site (as in Figure 2 A)<sup>[11]</sup> or b) aiding the recognition



**Figure 1.** The cation– $\pi$  interaction. A) Molecular electrostatic potential surface (negative potential in red, positive in blue) illustrating the electrostatic basis for a cation– $\pi$  interaction between benzene and an ammonium cation. Adapted with permission from Ref. [1c]. Copyright 2013 American Chemical Society. B) Examples of different interaction modes available to small-molecule systems engaging in cation– $\pi$  interactions (red dotted lines).

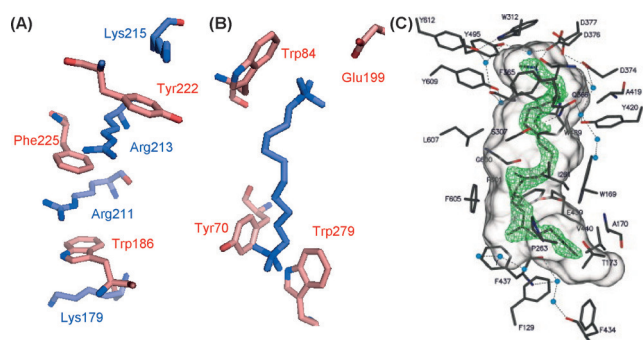
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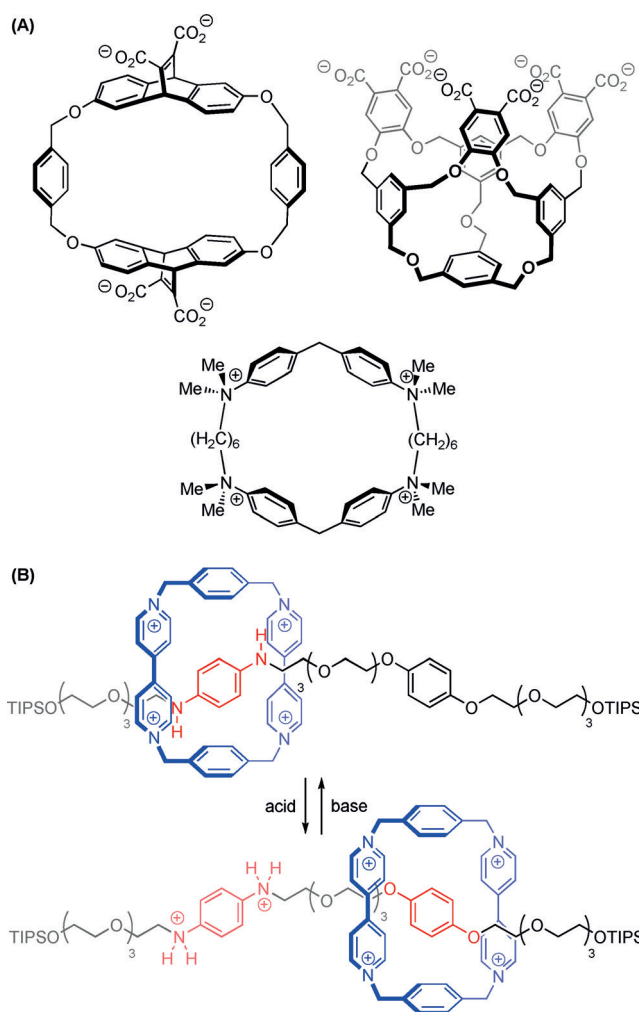


**Figure 2.** Examples of cation– $\pi$  interactions in biological systems. A) Extended cation– $\pi$  interactions in the human growth hormone receptor extracellular domain (Ref. [11e]). B) Crystal structure of acetylcholine esterase bound to a bis “quat” salt (Ref. [12a]). C) Crystal structure of squalene–hopene cyclase bound to a transition-state analogue for the cationic polycyclization of squalene. Adapted with permission from Ref. [13e]. Copyright 2004 Elsevier.

and binding of a ligand or substrate molecule in such an active site (as in Figure 2B).<sup>[12]</sup> In the latter case, the cation– $\pi$  interactions exerted by an enzyme can also stabilize a transition state relative to the resting ground state of its precursor, imparting rate acceleration and kinetic control of the selectivity in a chemical transformation (as in Figure 2C).<sup>[13]</sup>

Both the fundamental studies on the thermodynamics of the cation– $\pi$  interaction and the growing documentation of its importance in biological systems helped frame the potential for its application as a design element in synthetic chemistry.<sup>[14]</sup> Recognizing that the complementary strength and directionality of the cation– $\pi$  interaction could drive the association of small molecules into organized assemblies, researchers sought to utilize this attractive interaction for molecular recognition,<sup>[15]</sup> crystal engineering,<sup>[16]</sup> and materials design.<sup>[17]</sup> As a result, many of the early insights into the nuances of the cation– $\pi$  interaction arose from work in these fields (Scheme 1).<sup>[18]</sup>

The thorough characterization of the cation– $\pi$  interaction in enzymatic function and molecular recognition served to lay the foundation for its incorporation into the mechanistic understanding and design of small-molecule catalysts for organic synthesis.<sup>[1,13]</sup> In this Review, we analyze the burgeoning role of the cation– $\pi$  interaction in small-molecule

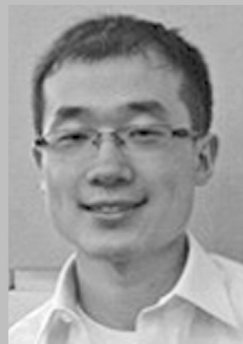


**Scheme 1.** Cation– $\pi$  interactions in molecular recognition. A) Select small-molecule cation acceptors (Ref. [15a–c]). B) A switchable molecular shuttle based on cation– $\pi$  interactions (Ref. [15d]). TIPS = triisopropylsilyl.

catalysis, placing particular emphasis on the elucidation of mechanisms by which such interactions govern the reactivity and/or selectivity of reactions. The well-established area of cationic transition-metal  $\pi$ -complexes will not be included,<sup>[19]</sup> as the transition metal– $\pi$  interaction possesses significant



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covalent character, which renders it distinct from the non-covalent cation– $\pi$  interaction. The systems described here are organized into three categories based on the role of the cation– $\pi$  interaction(s):

1. Substrate–substrate cation– $\pi$  interactions enable the organization of the reacting components. These interactions may occur intermolecularly to arrange multiple reacting partners relative to each other or intramolecularly to dictate substrate conformation. In this scenario, the catalyst does not engage directly in the cation– $\pi$  interaction.
2. Catalyst–catalyst cation– $\pi$  interactions control the catalyst conformation. The intramolecular cation– $\pi$  interaction between different motifs within a catalyst can help define the specific three-dimensional structure responsible for catalyst function. The cationic component in such an interaction may be native to the catalyst structure or may result from the formation of a covalent adduct with the substrate. In general, these intra-catalyst cation– $\pi$  interactions are analogous to those that influence the tertiary structures of biological macromolecules, as noted above.
3. Catalyst–substrate cation– $\pi$  interactions enhance the association of a catalyst with its substrate. These intermolecular interactions may exist in the ground state of a reactant–catalyst complex, or may serve to preferentially stabilize the transition structure of a reaction. These interactions may be further categorized into those in which the catalyst serves as a) the  $\pi$ -acceptor (i.e., the cation) or b) the  $\pi$ -donor. Cation– $\pi$  interactions of this type are again analogous to the enzyme active-site interactions identified in biological systems.

## 2. Substrate–Substrate Cation– $\pi$ Interactions

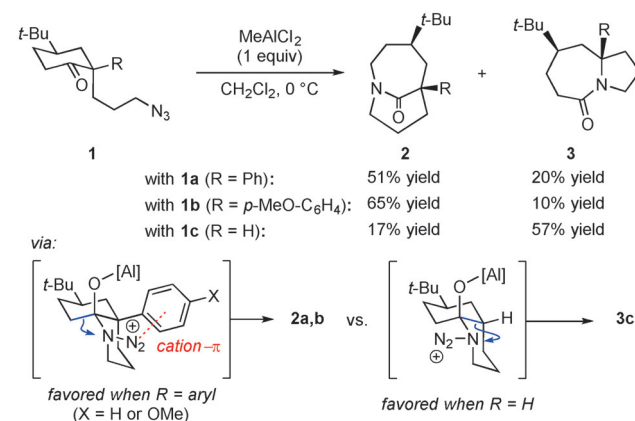
Organic small molecules bearing both a cationic component and a neutral aromatic residue tend to pack in the solid state with the two motifs in close spatial proximity.<sup>[20]</sup> The cation– $\pi$  interactions governing such packing can occur either between neighboring molecules or in an intramolecular fashion. This type of preorganization has been exploited to enable regioselective bond formation in solid-phase reactions.<sup>[21]</sup> Whereas similar principles can be engineered into solution-phase reactions, the cation– $\pi$  interactions driving

selective substrate preorganization must compete effectively with solvation effects. This Section reviews the application of this approach in solution-phase small-molecule-catalyzed or -promoted reactions.

### 2.1. Intramolecular Cation– $\pi$ Interactions

#### 2.1.1. Lewis Acid Promoted Regioselective Schmidt Annulation Reactions

Bridged bicyclic lactams bearing bridgehead nitrogen atoms are of considerable synthetic and structural interest owing to the exceptional reactivity of the distorted amide functional group.<sup>[22]</sup> In an elegant approach to accessing these motifs, Aubé and co-workers developed a Lewis acid promoted method for the intramolecular Schmidt reaction of alkyl azide substituted cyclohexanones (**1**).<sup>[23]</sup> As depicted in Scheme 2, alkyl azide addition to the activated carbonyl group is followed by C–C bond migration to afford either of two constitutional isomers: bridged bicyclic lactam **2** or fused bicyclic lactam **3**.



**Scheme 2.** Lewis acid promoted regioselective Schmidt annulation reactions. Blue arrows indicate C–C bond migration. The red dotted line indicates a cation– $\pi$  interaction.

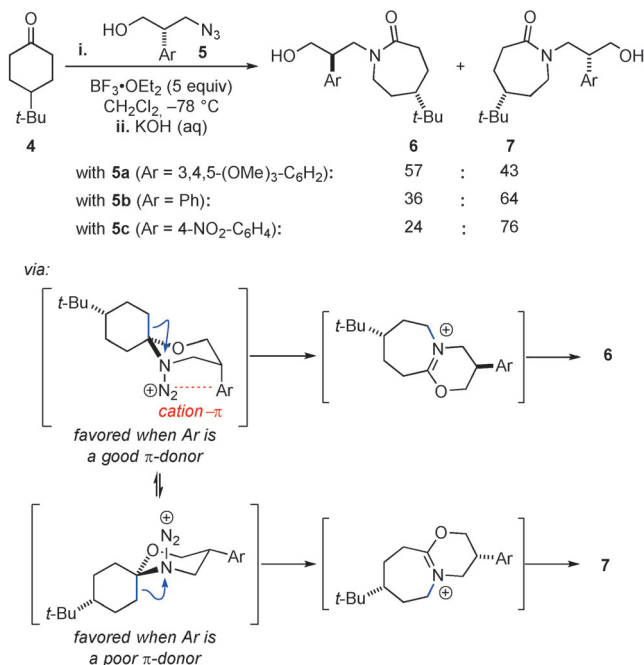
For 4-*tert*-butyl-substituted cyclohexanones, which react through a locked *trans*-decalin-like conformation, the regioselectivity of the bond migration was shown to be controllable through careful substrate design. A  $\pi$ -system installed at the  $\alpha$ -position (as in **1a** and **1b**) is appropriately situated to stabilize the intermediate, placing the diazonium cation in the axial position via a cation– $\pi$  interaction. Vicinal C–C bond migration from this conformation yields the desired bridged product **2**. However, the unsubstituted substrate **1c**, which lacks the potential to engage in this cation– $\pi$  interaction, reacts to form predominantly the fused side product **3**. Computational investigations of this system using density functional theory (DFT) corroborated the importance of such cation– $\pi$  interactions in controlling the regioselectivity of the bond-migration step.<sup>[24]</sup>



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### 2.1.2. Lewis Acid Promoted Diastereoselective Schmidt Ring-Expansion Reactions

Aubé and co-workers extended this strategy with a detailed study of the Lewis acid promoted diastereoselective Schmidt-type ring expansion of 4-*tert*-butyl-cyclohexanone (**4**) to form seven-membered lactams (**6** and **7**; Scheme 3).<sup>[25]</sup> In this transformation, the oxocarbenium ion



**Scheme 3.** Lewis acid promoted diastereoselective Schmidt ring-expansion reactions. Blue arrows indicate C–C bond migration. The red dotted line indicates a cation– $\pi$  interaction.

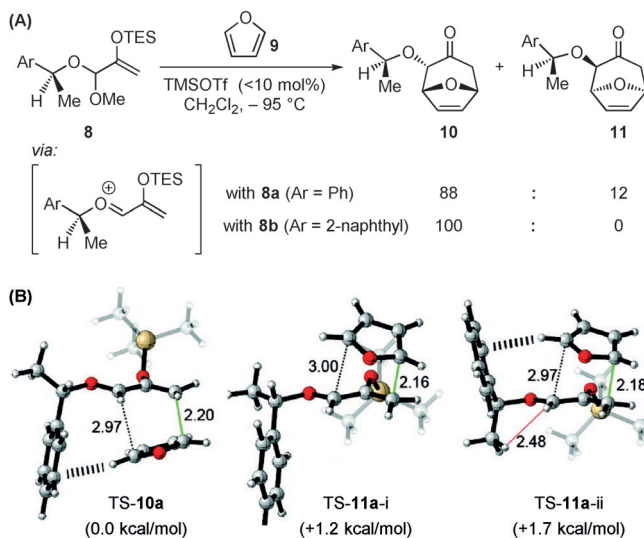
intermediate formed from the condensation of **4** and chiral hydroxyalkyl azide **5** undergoes nucleophilic attack by the pendant azide to form either of two diastereomeric diazonium ion intermediates. Only one of the two situates the diazonium ion in a position to engage in a cation- $\pi$  interaction with the aryl group. This interaction stabilizes the diazonium ion in an otherwise unfavorable axial conformation, from which C-C bond migration affords diastereomer **6**. Ring expansion from the equatorial conformation affords diastereomer **7**. The proposed role of this attractive interaction in stereodifferentiation is corroborated by the observation that the product diastereomeric ratio depends on the arene substitution, with the most electron-rich arenes affording the highest **6/7** ratios.

## 2.2. Intermolecular Cation- $\pi$ Interactions

### 2.2.1. Lewis Acid Catalyzed Diastereoselective [4+3] Cycloaddition Reactions

Lewis acid catalyzed [4+3] cycloaddition reactions of alkoxy-substituted siloxyallyl cations with electron-rich 1,3-dienes, such as furan, have long been successfully applied to

the efficient generation of substituted [3.2.1]-tricyclic scaffolds with good regio- and diastereoselectivity.<sup>[26]</sup> Hoffmann and co-workers demonstrated that chiral 2-arylethyl auxiliaries can induce high levels of stereocontrol in cycloadditions to afford enantiomerically pure 2-alkoxy-8-oxabicyclo[3.2.1]oct-6-en-3-ones (Scheme 4A).<sup>[27]</sup> For example, Lewis



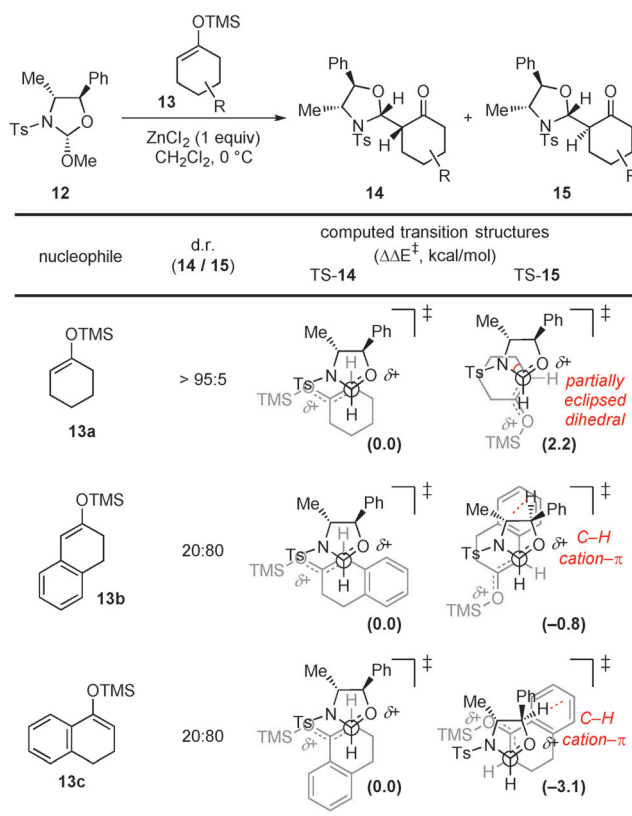
**Scheme 4.** A) Lewis acid catalyzed [4+3] cycloaddition and B) a stereochemical model for the observed diastereoselectivity. Black hashed lines indicate cation- $\pi$  interactions, while the red line represents a steric repulsion. Black and green dotted lines show the forming C-C bonds, with bond distances indicated in Å. The differential enthalpies of activation are shown in kcal mol<sup>-1</sup> (B3LYP/6-31G(d)). C gray, H white, O red, Si gold. Adapted with permission from Ref. [28a]. Copyright 2010 American Chemical Society. OTf = trifluoromethanesulfonate, TES = triethylsilyl, TMS = trimethylsilyl.

acid promoted cycloaddition of furan (**9**) with acetal **8a**, which bears a 1-phenylethanol-derived auxiliary, affords a mixture of the *endo* cycloadducts **10a** and **11a** in 88:12 d.r. The use of a 1-(2-naphthyl)ethanol-derived auxiliary bearing a more extended aryl group (as in **8b**) affords **10b** as a single diastereomer.

Computational analysis shed light on the origin of the observed diastereoselectivity (Scheme 4B).<sup>[28]</sup> In the lowest-energy computed transition structure (TS-**10a**), the chiral auxiliary is oriented to allow for an edge-to-face contact between the phenyl ring and the incoming furan (ca. 3 Å). This contact reflects an attractive cation- $\pi$  interaction between the  $\pi$ -face of the auxiliary and the partial positive charge developing at the 2-position of the furan nucleophile in the first transition state of the stepwise addition. In the two lowest-energy transition structures en route to minor diastereomer **11a**, this cation- $\pi$  stabilization is either absent (TS-**11a-i**) or accompanied by repulsive steric interactions (TS-**11a-ii**). Furthermore, this computational model reproduces the improved diastereoselectivity that was observed experimentally with the 1-(2-naphthyl)ethanol-derived auxiliary, reflecting the strengthened cation- $\pi$  interaction that results from the increased expanse of the  $\pi$ -donor.

### 2.2.2 Lewis Acid Catalyzed Diastereoselective Additions of Silyl Enol Ethers to a Chiral Oxazolinium Ion

Lewis acid catalyzed nucleophilic additions to enantioenriched oxazolinium ions offer an efficient method for formal stereoselective formylations of silyl enol ethers and allyl silanes.<sup>[29]</sup> Over the course of developing conditions for this transformation, Hoppe and co-workers found that a pseudoe-phedrine-derived oxazolidine (**12**) reacts with silyl enol ethers to afford  $\alpha$ -substituted ketones. Remarkably, silyl enol ethers bearing aromatic substituents are alkylated with the opposite sense of diastereoselectivity relative to simpler aliphatic analogues (**14** vs. **15**, Scheme 5).<sup>[29c]</sup>



**Scheme 5.** Lewis acid promoted diastereoselective additions of silyl enol ethers to a chiral oxazolinium ion and computational stereochemical models for the observed diastereoselectivity. Red dotted lines indicate attractive cation– $\pi$  interactions, while the solid red curve represents the dihedral angle leading to repulsive steric interactions. Black and gray dotted lines show the breaking and forming bonds. The differential energies of activation are shown in kcal mol<sup>-1</sup> (M06-2X/6-311 + G(d,p)//B3LYP/6-31G(d)).

While the diastereoselective formation of **14a** may be rationalized simply on the basis of steric repulsion between the nucleophile and the electrophile in the minor transition state, computational analysis provided insight regarding the basis of the selectivity with aromatic substrates.<sup>[30]</sup> Computational models utilizing density functionals corrected to account for dispersion interactions predict the correct sense of diastereoselectivity for both classes of silyl enol ethers

(e.g., **13a–13c**). The lowest-energy computed transition structures en route to the aromatic products **15b** and **15c** exhibit van der Waals contacts (2.71–2.92 Å) between the substrate arenes and the  $\alpha$ -C–H bond of the oxazolinium ion, which bears a partial positive charge. These cation– $\pi$  interactions override the energetic cost of distortion to avoid steric interactions in the stacked transition structure. Consequently, the reversal of diastereoselectivity observed in reactions with silyl enol ethers **13b** and **13c** is attributed to stabilizing cation– $\pi$  interactions in the transition structure.

### 3. Catalyst–Catalyst Cation– $\pi$ Interactions

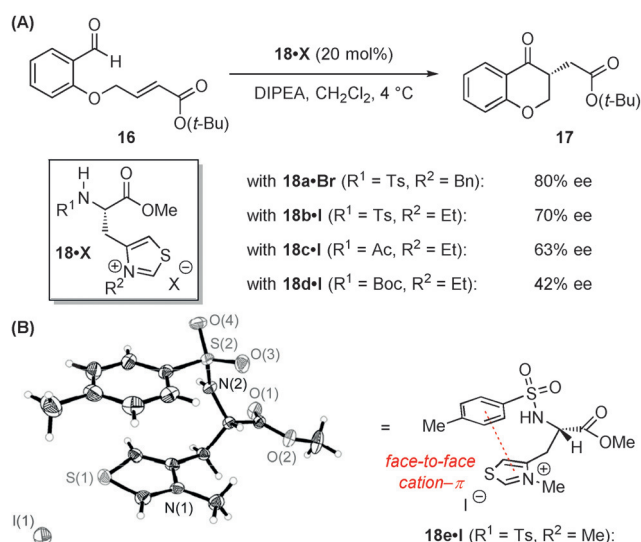
As noted in Section 1, structural biologists have found that evidence for cation– $\pi$  interactions is ubiquitous among the structures in the Protein Data Bank. For example, when a cationic amino acid side chain (e.g., Lys, Arg) is positioned near an aromatic side chain (e.g., Phe, Tyr, Trp), the global protein structure is often biased to enable effective cation– $\pi$  interactions between the residues.<sup>[1,11]</sup> Along with hydrogen bonding, salt-bridge formation, hydrophobic packing, and disulfide formation, such cation– $\pi$  interactions between amino acid residues play a major role in determining the tertiary and quaternary structures of proteins, which in turn govern their functions. The spatial organization of small-molecule catalysts can be guided in a similar manner by cation– $\pi$  interactions between discrete functional groups within the catalyst framework.<sup>[31]</sup> The cationic component in such an interaction may be native to the catalyst structure or may result from the formation of a covalent adduct with a substrate in situ. This Section reviews several cases in which one or more intramolecular cation– $\pi$  interactions alter the conformation of the catalyst active site to influence reactivity and/or selectivity.

#### 3.1. Native Cationic Catalysts

##### 3.1.1. Peptidic Thiazolium Ion Catalysts for Stetter Reactions

Miller and co-workers evaluated a family of short peptidic thiazolium catalysts for the asymmetric Stetter cyclization of salicylaldehyde-derived enoates (**16**; Scheme 6A).<sup>[32]</sup> Simple  $\beta$ -thiazolyl alanine derivatives were found to be among the most effective catalysts, with catalyst **18a** affording cyclization product **17** in up to 80% *ee*. Extensive studies on the effect of the catalyst structure demonstrated that the enantioselectivity is highly dependent on the identity of the N substituent of the catalyst. When the *para*-toluenesulfonyl group (Ts) of catalyst **18b** is replaced with an acetyl (Ac; catalyst **18c**) or *tert*-butoxycarbonyl (Boc; catalyst **18d**) group, cyclization proceeds with markedly diminished enantioselectivity.

X-ray crystallographic analysis of **18e** revealed that an intramolecular cation– $\pi$  interaction between the aromatic ring of the Ts substituent and the thiazolium ion favors a folded conformation (Scheme 6B), which is proposed to be maintained in the cationic Breslow intermediate involved in the key C–C bond-forming step of the catalytic cycle.



**Scheme 6.** A) Asymmetric Stetter reaction catalyzed by  $\beta$ -thiazolyl alanine derivatives. B) X-ray crystal structure and line representation of such a catalyst. Red dotted lines indicate cation– $\pi$  interactions. Adapted from Ref. [32] with permission of the Royal Society of Chemistry. DIPEA = *N,N*-diisopropylethylamine.

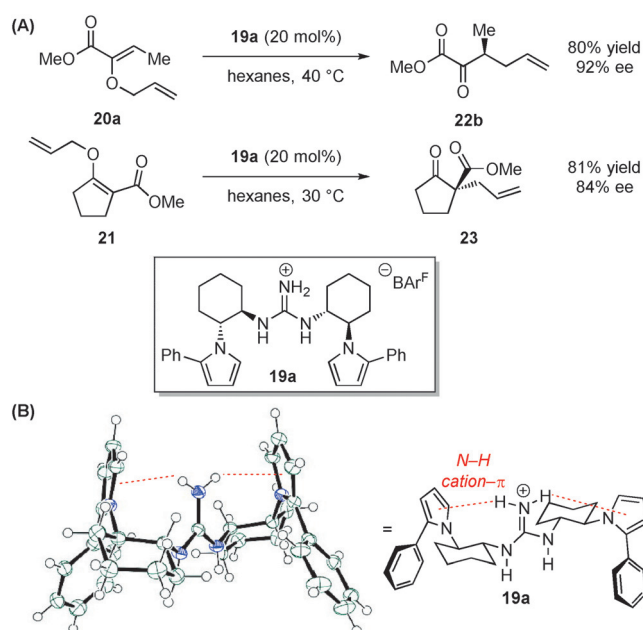
Accordingly, the effective enantioinduction is attributed to the stereochemically well-defined environment shaped by this cation– $\pi$  interaction.

### 3.1.2. Guanidinium Ion Catalysts for Claisen Rearrangements

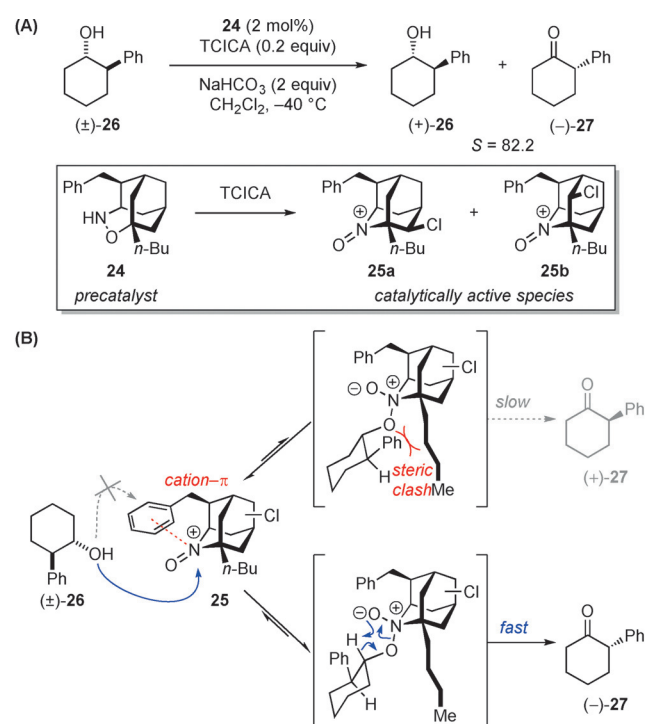
Jacobsen and co-workers disclosed a series of C<sub>2</sub>-symmetric guanidinium ion catalysts **19** for enantioselective Claisen rearrangements of  $\alpha$ - and  $\beta$ -ketoesters (**20** and **21**; Scheme 7A).<sup>[33]</sup> X-ray structural analysis, solution-phase ROESY studies, and gas-phase computations all indicated that the ground-state catalyst conformation positions the guanidinium NH<sub>2</sub> hydrogen atoms in close contact with the  $\pi$ -faces of the proximal pyrrole rings (3.23 Å in the calculated structure, 3.22 and 3.19 Å in the crystal structure; Scheme 7B). These cation– $\pi$  interactions serve to rigidify the catalyst conformation and position the aryl substituents of the pyrroles appropriately to participate in stereodifferentiating interactions in the rearrangement transition state. The nature of these secondary interactions will be discussed in Section 4.2.6.

### 3.1.3. Oxoammonium Ion Catalysts for Alcohol Oxidations

Iwabuchi and co-workers advanced a method employing adamantane-derived alkoxyamine precatalyst **24** for highly stereoselective oxidative kinetic resolution (OKR) reactions of unactivated secondary alcohols (Scheme 8A).<sup>[34]</sup> This method was based on prior work utilizing the same family of precatalysts for simple oxidations and the kinetic resolution of activated alcohols.<sup>[35]</sup> On the basis of careful spectroscopic analysis and independent isolation experiments, the active catalytic species was identified as a pair of chlorinated oxoammonium ions (**25a** and **25b**) that are generated from the reaction of **24** with trichloroisocyanuric acid (TCICA).



**Scheme 7.** A) Guanidinium ion catalyzed asymmetric Claisen rearrangements. B) X-ray crystal structure and line representation of the guanidinium ion catalyst. Red dotted lines indicate cation– $\pi$  interactions. Solvent molecules omitted for clarity. Adapted with permission from Ref. [33c]. Copyright 2011 American Chemical Society.



**Scheme 8.** A) Alkoxyamine-catalyzed kinetic resolution of secondary alcohol **26**. B) A stereochemical model for the observed enantiodiscrimination. The red dotted line indicates an attractive cation– $\pi$  interaction, while the red curves indicate a repulsive steric interaction. TCICA = trichloroisocyanuric acid.

In the proposed mechanism, the benzyl substituent in either active catalyst (**25a** or **25b**) forms an intramolecular cation– $\pi$  interaction with the oxoammonium ion, thereby shielding one face as the alcohol substrate (**26**) adds reversibly to the exposed face to form a covalent adduct (Scheme 8B). Stereodifferentiation with a selectivity factor as high as  $S = 296$  was achieved in the subsequent oxy-Cope-type elimination through differential steric interactions with the  $\alpha$ -alkyl group. In this example, the cation– $\pi$  interaction is deemed to play a critical role in the overall kinetic resolution process by enforcing the diastereoselective formation of the initial covalent intermediate.

### 3.2. Cationic Catalyst States Formed On-Cycle

#### 3.2.1. Nucleophilic Catalysts for Acyl-Transfer Reactions

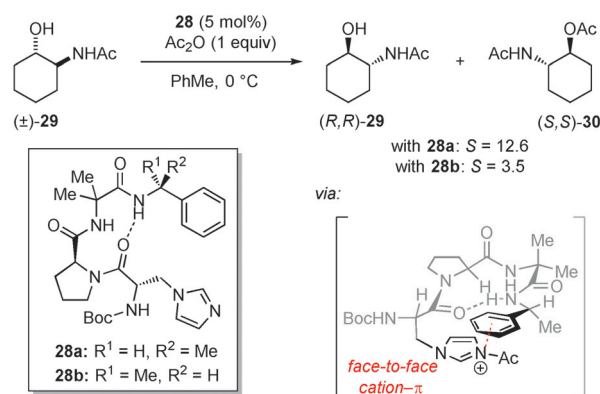
Nucleophilic heterocycles, and pyridine derivatives in particular, have been studied extensively as efficient catalysts for a range of acyl-transfer reactions.<sup>[36]</sup> In the generally accepted mechanism, these nucleophilic catalysts undergo N acylation to form highly electrophilic, cationic, covalent adducts prior to acyl transfer to a substrate nucleophile. In principle,  $\pi$ -donating groups could be integrated into the scaffolds of these catalysts to enable the cation– $\pi$  stabilization of specific conformations of the acylated intermediate, potentially leading to improved reactivity or selectivity. The viability of this strategy has been demonstrated in several systems, as described below.

##### 3.2.1.1. Peptidic N-Alkylimidazole Catalysts

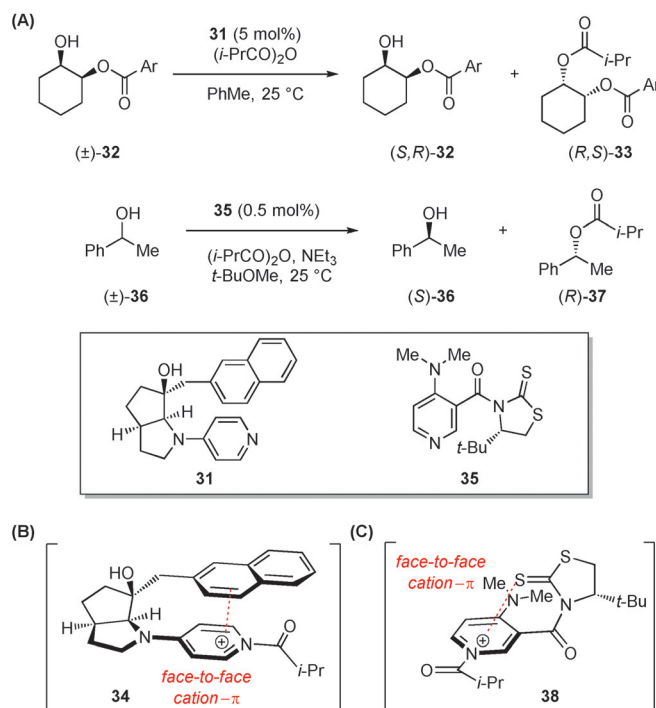
Inspired by the catalytic efficiency of *N*-methylimidazole (NMI) in alcohol acylations, Miller and co-workers sought to develop structurally analogous chiral peptidic catalysts for asymmetric acylation reactions. Towards this aim, they identified  $\beta$ -imidazolyl alanine containing tripeptide catalyst **28a** for the kinetic resolution of acetamide-substituted secondary alcohols (such as **29**; Scheme 9).<sup>[37]</sup> The C-terminal  $\alpha$ -methylbenzylamide moiety was incorporated to allow for an intramolecular cation– $\pi$  interaction with the charged acyl imidazolium ion intermediate. In line with the hypothesized role, the selectivity factor of the kinetic resolution of model secondary alcohol **29** was found to be highly sensitive to the configuration of the  $\alpha$ -methylbenzylamide moiety; diastereomeric catalyst **28b** afforded acylated product **30** with significantly reduced stereoselectivity ( $S = 3.5$  with **28b** compared to  $S = 12.6$  with **28a**). Although the precise mechanistic role of the  $\alpha$ -methylbenzylamide group has not been ascertained, its presumed participation in a cation– $\pi$  interaction has served as the basis for further catalyst development.<sup>[37b]</sup>

##### 3.2.1.2. Chiral Aminopyridine Catalysts

Fuji and co-workers disclosed a naphthyl-substituted pyrrolidinopyridine catalyst for the acylative kinetic resolution of secondary alcohols (Scheme 10A).<sup>[38]</sup> Nuclear Overhauser effect (nOe) experiments with the parent (**31**) and acylated (**34**) catalysts revealed that acylation induces a con-



**Scheme 9.** Tripeptide-catalyzed acylative kinetic resolution of acetamide-substituted secondary alcohols. The red dotted line indicates a putative cation– $\pi$  interaction. Boc = *tert*-butoxycarbonyl.



**Scheme 10.** A) Acylative kinetic resolution of secondary alcohols catalyzed by chiral pyridines. B, C) Line representation of the stacked conformations adopted by the *N*-acylpyridinium catalyst adducts. Red dotted lines indicate cation– $\pi$  interactions.

formational change driven by an intramolecular cation– $\pi$  interaction between the pendent naphthyl group and the pyridinium ion (Scheme 10B).

Yamada et al. explored a related conformation-switching strategy in the application of a chiral thiazolidinethione-functionalized *N,N*-dimethyl-4-aminopyridine (DMAP) catalyst (**35**) to the acylative kinetic resolution of secondary benzylic alcohols (**36**; Scheme 10A).<sup>[39]</sup> X-ray structural analysis, NMR characterization, and UV/Vis spectroscopic studies all indicated that **35** also undergoes a conformational change upon acylation, which is driven by an intramolecular cation– $\pi$  interaction between the thiocarbonyl group and the

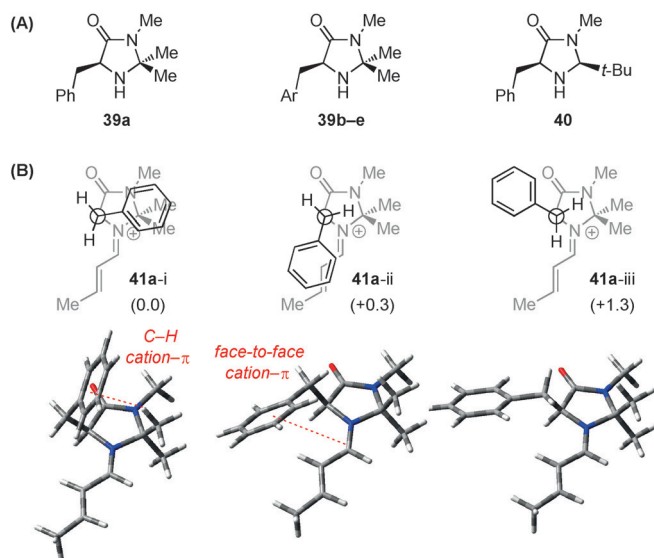
pyridinium ion (Scheme 10C).<sup>[40]</sup> Computational (MP2) studies corroborate the role of the cation– $\pi$  interaction in the conformation-switching properties of catalysts **31** and **35**.<sup>[41]</sup>

In both cases, the chiral environment defined by the cation– $\pi$  interaction controls the  $\pi$ -facial reactivity of the *N*-acylpyridinium intermediate in enantioselective acyl transfer to alcohol substrates. The differential recognition of enantiomeric substrates is purportedly assisted by additional intermolecular cation– $\pi$  interactions between the aromatic group on the substrate and the pyridinium ion, as described further in Section 4.1.3. Similar catalytic conditions enable the acylative dynamic kinetic resolution of hemiaminals,<sup>[42]</sup> and an analogous chiral pyridine auxiliary has been applied to two-step enantioselective cyclopropanation reactions.<sup>[43]</sup>

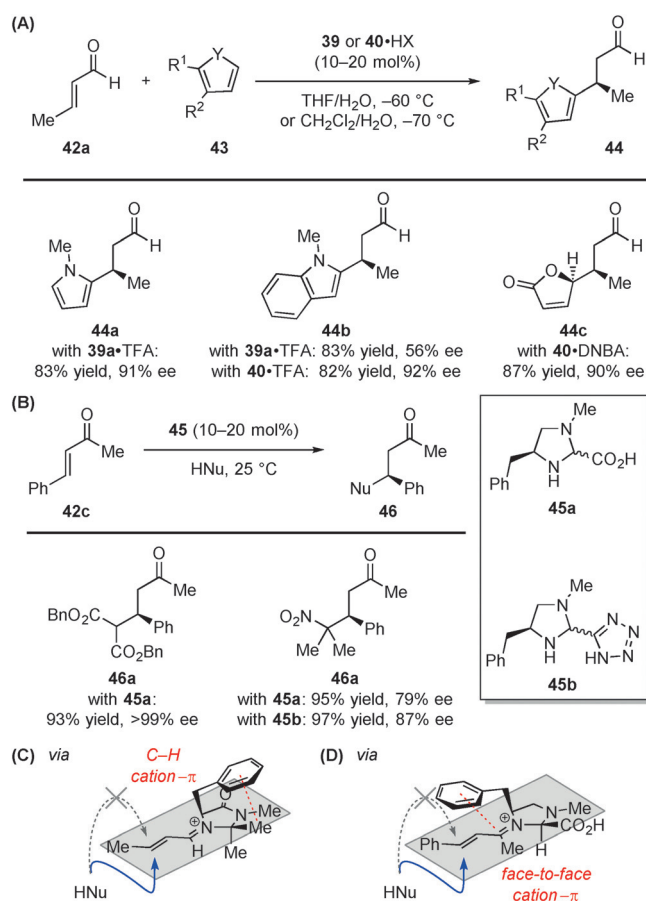
### 3.2.2. Secondary Amine Catalysts for the Activation of Enals

Following their introduction by MacMillan and co-workers, chiral imidazolidinone derivatives, such as **39** and **40** (see Figure 3 A), have been utilized widely to promote catalytic enantioselective transformations of  $\alpha,\beta$ -unsaturated aldehydes and ketones via iminium ion intermediates (Scheme 11 A).<sup>[44]</sup> The groups of MacMillan,<sup>[45]</sup> Houk,<sup>[46]</sup> and Yamada<sup>[47]</sup> independently undertook computational studies of these catalytic systems in an effort to elucidate the seemingly general basis for enantioinduction across a broad range of reactions. Using molecular mechanics and DFT to evaluate the accessible ground-state structures of the preferred *E*-iminium adduct (**41a**), three low-energy conformations differing in their Ph–C–C–N dihedral angles were identified (Figure 3).

Taken together, conformations **41a-i** and **41a-ii**, both of which position the phenyl ring over regions of the iminium ion



**Figure 3.** A) Selected chiral imidazolidinone catalysts. B) Lowest-energy conformations of the iminium ion intermediate formed from **39a** and crotonaldehyde. Relative energies (kcal mol<sup>−1</sup>) predicted by DFT calculations (B3LYP/6-31 g(d)) are shown in parentheses. Red dotted lines indicate cation– $\pi$  interactions. C gray, H white, N blue, O red.



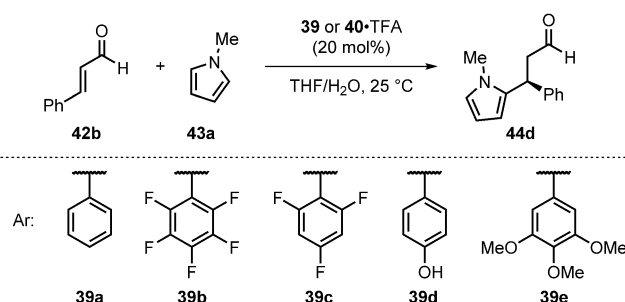
**Scheme 11.** A) Select imidazolidinone- and B) imidazoline-catalyzed Michael reactions. C, D) Stereochemical models for the stereoselective nucleophile addition to the activated iminium ion intermediates. Red dotted lines indicate cation– $\pi$  interactions. Blue arrows indicate favorable trajectories for nucleophile approach, while gray dotted arrows indicate sterically blocked trajectories. DNBA = 3,5-dinitrobenzoic acid, TFA = trifluoroacetic acid.

that bear positive charge, are predicted to account for 85 % of all possible species at 25 °C.<sup>[46a]</sup> Further ab initio (MP2) energy decomposition analysis predicts that the electrostatic and polarization terms dominate the attractive interactions between the phenyl ring and the iminium cation.<sup>[47]</sup> Conformation **41a-iii**, in which the phenyl ring is turned away from the face of the iminium ion, cannot partake in such stabilizing interactions and is accordingly markedly higher in energy. Iminium ions derived from *tert*-butyl-substituted catalyst **40** show similar conformational preferences. With either catalyst, the cation– $\pi$  interaction enables the phenyl ring to effectively shield the *Si* face of the iminium ion intermediate, thereby favoring nucleophilic addition to the *Re* face of this electrophile (Scheme 11 C). This facial preference and the preservation of the cation– $\pi$  interaction in the C–C bond-forming step are further reflected in the computed transition-state energies.

Jørgensen and co-workers introduced related imidazoline catalysts (**45a** and **45b**) for conjugate additions to  $\alpha,\beta$ -unsaturated ketones (Scheme 11 B).<sup>[48]</sup> Based on computational modeling (PM3), the authors suggested that these

catalysts achieve enantiocontrol through cation– $\pi$  interactions analogous to those described above. The benzyl substituent of the imidazolidinone catalyst shields the *Re* face of the iminium ion intermediate, thereby favoring nitronate or enolate addition from the *Si* face of the electrophile (Scheme 11D).

To rigorously evaluate this stereochemical model, Gilmour and co-workers examined a series of imidazolidinone catalysts (**39a–39e**) differing only in their aryl substituents.<sup>[49]</sup> In a model reaction, namely the conjugate addition of *N*-methylpyrrole to cinnamaldehyde (**42b**), the observed enantioselectivity was shown to display a strong dependence on the arene quadrupole moment ( $Q_{ZZ}$ ; Scheme 12, Table 1). X-ray structural analysis, supported by DFT calculations, further revealed that the ground-state distribution of the iminium ion conformations (**41x-i–41x-iii** where **x = a–e**) is



**Scheme 12.** Asymmetric conjugate addition reactions catalyzed by chiral imidazolidinones bearing electronically varied aryl substituents.

**Table 1:** Reaction enantioselectivity as a function of the electronic properties of the aromatic group on the catalyst.

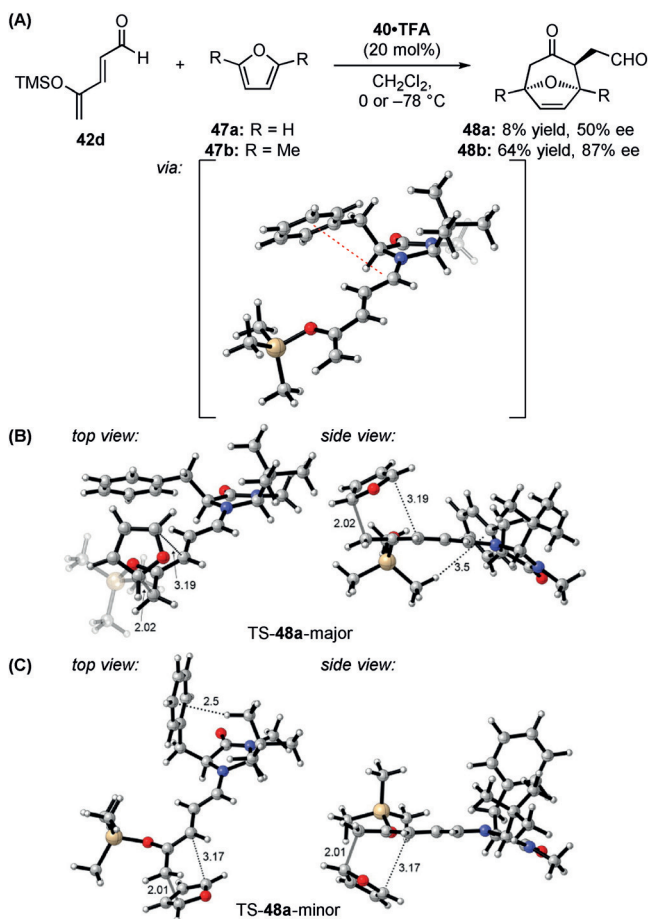
| Entry | Catalyst <sup>[a]</sup> | $Q_{ZZ}$ | Iminium ion mole fraction |               |                | <i>ee</i> <sup>[b]</sup> [%] |
|-------|-------------------------|----------|---------------------------|---------------|----------------|------------------------------|
|       |                         |          | <b>41x-i</b>              | <b>41x-ii</b> | <b>41x-iii</b> |                              |
| 1     | <b>39b</b>              | +3.01    | 0.37                      | 0.16          | 0.47           | 65                           |
| 2     | <b>39c</b>              | +0.28    | 0.39                      | 0.21          | 0.40           | 70                           |
| 3     | <b>39a</b>              | −3.46    | 0.75                      | 0.03          | 0.23           | 84                           |
| 4     | <b>39d</b>              | −3.71    | 0.76                      | 0.04          | 0.20           | 90                           |
| 5     | <b>39e</b>              | −5.68    | 0.65                      | 0.17          | 0.18           | 94                           |
| 6     | <b>40</b>               | −3.46    | –                         | –             | –              | 85                           |

[a] See the catalyst substitution pattern in Scheme 12. [b] Enantiomeric excess of **44d** formed by the reaction shown in Scheme 12.

affected by the electronic properties of the aryl group. Catalysts bearing electron-rich arenes favor conformers **41x-i** and **41x-ii**, whereas catalysts bearing electron-deficient arenes display a preference for the splayed conformers **41x-iii**. Taken together, these results offer compelling evidence that attractive cation– $\pi$  interactions play an important role in the stereochemical organization of the reactive iminium ion intermediate, thereby enabling highly selective nucleophilic addition.

In an independent effort, Harmata and co-workers developed a method for the formal [4+3] cycloaddition

reaction of 4-trialkylsilyloxypentadienal **42d** with furan nucleophiles **47** (Scheme 13).<sup>[50]</sup> Remarkably, the sense of enantioinduction observed with catalyst **40** is opposite to that anticipated from analyses of MacMillan's [4+2] and conjugate addition systems.<sup>[51]</sup> Therefore, further computational studies were undertaken to rationalize this deviation from known models.<sup>[52]</sup>



**Scheme 13.** A) Imidazolidinone-catalyzed [4+3] cycloaddition and the lowest-energy computed iminium ion conformation. The lowest-energy computed transition structures (M06-2X/6-311+G(d,p)//B3LYP/6-31G(d)) en route to the B) major and C) minor enantiomers of **48**. The red dotted line indicates a cation– $\pi$  interaction, and black dotted lines indicate other noncovalent interactions. Faded gray lines indicate forming bonds. C gray, H white, N blue, O red, Si gold. Key distances shown in Å. Adapted from Ref. [52].

DFT calculations identified low-energy iminium ion conformations analogous to **41x-i–41x-iii**, wherein intramolecular cation– $\pi$  interactions favor positioning the phenyl ring over the iminium ion. However, as the enantioselectivity-determining C–C bond formation occurs at the most-distal  $\delta$ -position of the conjugated iminium ion, it was hypothesized that the furan nucleophile could approach either the *Si* or the *Re* face without steric penalty. However, furan approach requires rotating the trimethylsilyl group out of the plane of the iminium ion. Whereas rotation towards the *Re* face (opposite *Si* face furan approach) requires minimal reorgan-

ization of the benzyl and *tert*-butyl groups in the transition state (Scheme 13B), rotation toward the *Si* face induces significant distortion (Scheme 13C). The energetic cost for this transition-state distortion from the cation- $\pi$ -stabilized ground-state geometry was proposed to account for the chirality transfer in this unusual system. This example provides yet another illustration of how cation- $\pi$  interactions can exert long-ranging and subtle effects on catalyst conformation.

#### 4. Catalyst–Substrate Cation- $\pi$ Interactions

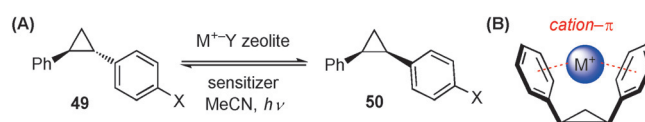
The importance of cation- $\pi$  interactions for substrate recognition and transition-state stabilization in enzymatic catalysis has been established in numerous studies.<sup>[10]</sup> In a quintessential example, the dense array of cation- $\pi$  interactions in the active site of squalene-hopene cyclase stabilizes charge redistribution in the transition state of the cationic polycyclization central to steroid biosynthesis.<sup>[13]</sup> Squalene-hopene cyclase and a host of other enzymes achieve remarkable rate acceleration and precise stereocontrol through the cooperative action of multiple amino acid residues, many of which contain charged or aromatic moieties, lining the active site. The elucidation of substrate-active site interactions in enzymatic systems has paved the way for the application of the cation- $\pi$  interaction as a design element for small-molecule catalysis. This Section reviews key examples in which cation- $\pi$  interactions have been invoked or demonstrated to play a key role in influencing reaction outcomes in synthetic catalyst systems.

##### 4.1. Substrate $\pi$ -Donors

##### 4.1.1. Alkali Metal Ion Controlled Diastereoselective Photoisomerization of Diphenylcyclopropane

The *trans* and *cis* isomers of diphenylcyclopropane (**49a** and **50a**) can undergo photoisomerization in the presence of an acetanilide sensitizer to form a thermodynamic mixture of the two diastereomers (55:45 *trans/cis*).<sup>[53]</sup> Ramamurthy and co-workers hypothesized that the introduction of alkali metal ions to the reaction mixture could perturb the equilibrium distribution to favor the *cis* isomer (**50a**), which would presumably bind the cation more tightly within the concave arrangement of its phenyl rings. They subsequently demonstrated that free alkali metal ions introduced into the reaction medium via a dry, metal-exchanged zeolite exert significant influence over the diastereomeric composition of the photostationary state (Scheme 14, Table 2).<sup>[54]</sup> Whereas all of the alkali metal ions ( $\text{Li}^+$ ,  $\text{Na}^+$ ,  $\text{K}^+$ ,  $\text{Rb}^+$ , and  $\text{Cs}^+$ ) favor the formation of *cis* isomer **50a**, the smaller ions ( $\text{Li}^+$  and  $\text{Na}^+$ ), which generally engage in stronger cation- $\pi$  interactions, afford the highest diastereomeric ratios.

The critical role of the cation- $\pi$  interaction in the stereocontrol of the photoisomerization was further substantiated by control experiments with monosubstituted diarylcyclopropanes (**49/50b, c**). Consistent with prior studies,<sup>[7]</sup> the computed cation-binding energies for *para*-methoxy-substi-



**Scheme 14.** A) Alkali metal ion promoted diastereoselective photoisomerization of diarylcyclopropanes. B) Schematic representation of the cation- $\pi$  interactions favoring formation of the *cis* isomer. Red dotted lines indicate cation- $\pi$  interactions.

**Table 2:** Perturbation of the photostationary state of diarylcyclopropanes with alkali metal cations.

| Entry | Substrate <sup>[a]</sup> | X   | M <sup>+</sup> | $E_{\text{int}}^{\text{[b]}}$<br>[kcal mol <sup>-1</sup> ] | 49/50 <sup>[c]</sup> |
|-------|--------------------------|-----|----------------|------------------------------------------------------------|----------------------|
| 1     | 49/50a                   | H   | –              | n/a                                                        | 44:55                |
| 2     |                          |     | $\text{Li}^+$  | –75.8                                                      | 91:9                 |
| 3     |                          |     | $\text{Na}^+$  | –53.9                                                      | 92:8                 |
| 4     |                          |     | $\text{K}^+$   | –32.6                                                      | 88:12                |
| 5     |                          |     | $\text{Rb}^+$  | –28.2                                                      | 85:15                |
| 6     |                          |     | $\text{Cs}^+$  | –22.7                                                      | 65:35                |
| 7     | 49/50b                   | OMe | $\text{Na}^+$  | –53.7                                                      | n.d.                 |
| 8     | 49/50c                   | CN  | –              | n/a                                                        | 45:55                |
| 9     |                          |     | $\text{Na}^+$  | –43                                                        | 44:46                |

[a] See the substrate structure and substitution pattern in Scheme 14.

[b]  $E_{\text{int}}$  is the computed interaction energy between **50** and  $\text{M}^+$ , calculated with MP2/6-31 g\*/Hay-Wadt ECP (K, Rb, Cs). [c] Isomer ratio at the photostationary state.

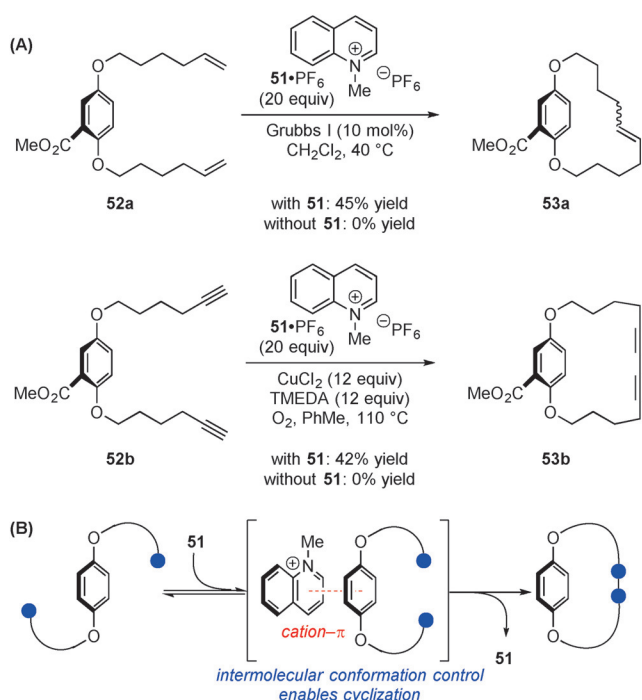
tuted **50b** are similar to those for the parent diphenylcyclopropane **50a**. However, *para*-cyano-substituted **50c** was calculated to be a much weaker  $\pi$ -donor, compromising its ability to participate in a templating cation- $\pi$  interaction with the metal ions. Accordingly, the photostationary diastereomeric ratio obtained with **50c** in the presence of  $\text{Na}^+$  (44:56 *trans/cis*) exhibits little deviation from that obtained in the absence of a template (45:55 *trans/cis*), highlighting the importance of electronic effects on the  $\pi$ -system in such interactions.

##### 4.1.2. Cation-Templated Cyclization Reactions

Large and medium-sized rings are pervasive among natural products and pharmaceuticals; however, cyclization reactions to form these structural motifs are generally quite challenging owing to the significant entropic and enthalpic costs of bringing remote termini into close proximity.<sup>[55]</sup> Some small rings can be equally challenging to form because of the precise stereoelectronic alignment required for bond formation. In any of these cases, the use of attractive interactions to preorganize the reactive functional groups prior to cyclization may, in theory, reduce the entropic barrier to cyclization and control the stereoelectronic alignment necessary for selective bond formation. Towards this aim, the cation- $\pi$  interaction has been applied as a design element to enable several challenging cyclization reactions.

## 4.1.2.1. Quinolinium Ion Promoted Macrocyclization Reactions

In an elegant demonstration of the application of cation– $\pi$  interactions to facilitate cyclization reactions, Collins and co-workers explored the use of pyridinium salts as intermolecular conformational-control elements for the macrocyclization of hydroquinone derivatives (**52**).<sup>[56]</sup> Whereas no detectable cyclization occurs in the absence of a template, upon introduction of quinolinium salt **51** to the reaction medium, cation– $\pi$  interactions enforce a substrate conformation that enables ring-closing metathesis or intramolecular Glaser–Hay coupling (Scheme 15). This intermolecular conformational-control strategy enables the preparation of cyclophanes **53** in 40–60% yield, even under relatively concentrated conditions. Similarly, pyridinium and tetraalkylammonium salts have been employed as templates to amplify the production of high-affinity macrocyclic hosts in the synthesis of dynamic combinatorial libraries.<sup>[57]</sup>

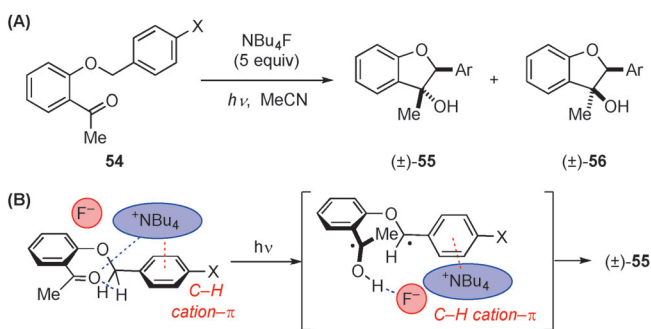


**Scheme 15.** A) Pyridinium-promoted macrocyclization reactions through B) conformational control enabled by a cation– $\pi$  interaction. The red dotted line indicates the cation– $\pi$  interaction. Blue circles indicate reactive functional groups. TMEDA = *N,N,N',N'*-tetramethylethylenediamine.

## 4.1.2.2. Tetraalkylammonium-Templated Norrish–Yang Cyclizations

In a related vein, Yamada et al. reported a method for the diastereoselective tetraalkylammonium-templated Norrish–Yang cyclization of 2-benzyloxy-acylbenzenes **54** to afford *trans*-dihydrobenzofuranols **55**.<sup>[58]</sup> In the absence of any additives, the photochemical cyclization of **54a** in trifluorotoluene yields exclusively *cis*-dihydrobenzofuranol **56a**. However, in the presence of an excess of the tetraalkylammonium

salt, this diastereoselectivity is overturned to favor the *trans*-dihydrobenzofuranol with a diastereomeric ratio as high as 93:7 (**55a/56a**). Furthermore, the degree of stereoselectivity depends on the  $\pi$ -donating ability of the benzyl group of the substrate (as in substrates **54a–54d**), with the most electron-rich substrates affording the highest *trans/cis* ratios (Scheme 16A, Table 3). The high degree of diastereomeric control was attributed to the cooperative effect of multiple attractive interactions between substrate and ammonium ion, including



**Scheme 16.** A) Tetrabutylammonium-templated Norrish–Yang cyclization and B) a model for the observed diastereoselectivity. Red dotted lines represent cation– $\pi$  interactions; blue dotted lines represent other noncovalent interactions.

**Table 3:** Tetrabutylammonium-templated Norrish–Yang cyclization.

| Entry | Substrate <sup>[a]</sup> | X               | Solvent           | NBu <sub>4</sub> F (equiv) | Conv. <sup>[b]</sup> [%] | d.r. ( <b>55/56</b> ) |
|-------|--------------------------|-----------------|-------------------|----------------------------|--------------------------|-----------------------|
| 1     | <b>54a</b>               | H               | PhCF <sub>3</sub> | 0                          | 17 <sup>[c]</sup>        | 0:100 <sup>[c]</sup>  |
| 2     |                          |                 |                   | 5                          | 93 <sup>[c]</sup>        | 93:7 <sup>[c]</sup>   |
| 3     |                          |                 | MeCN              | 0                          | 35 <sup>[c]</sup>        | 17:83 <sup>[c]</sup>  |
| 4     |                          |                 |                   | 5                          | 96 <sup>[c]</sup>        | 84:16 <sup>[c]</sup>  |
| 5     | <b>54b</b>               | Me              | MeCN              | 0                          | 35                       | 20:80                 |
| 6     |                          |                 |                   | 5                          | 84                       | 93:7                  |
| 7     | <b>54c</b>               | OMe             | MeCN              | 0                          | 70                       | 11:89                 |
| 8     |                          |                 |                   | 5                          | 99                       | 94:6                  |
| 9     | <b>54d</b>               | CF <sub>3</sub> | MeCN              | 0                          | 33                       | 23:77                 |
| 10    |                          |                 |                   | 5                          | 6                        | 59:41                 |

[a] See the substrate structure and substitution pattern in Scheme 16.

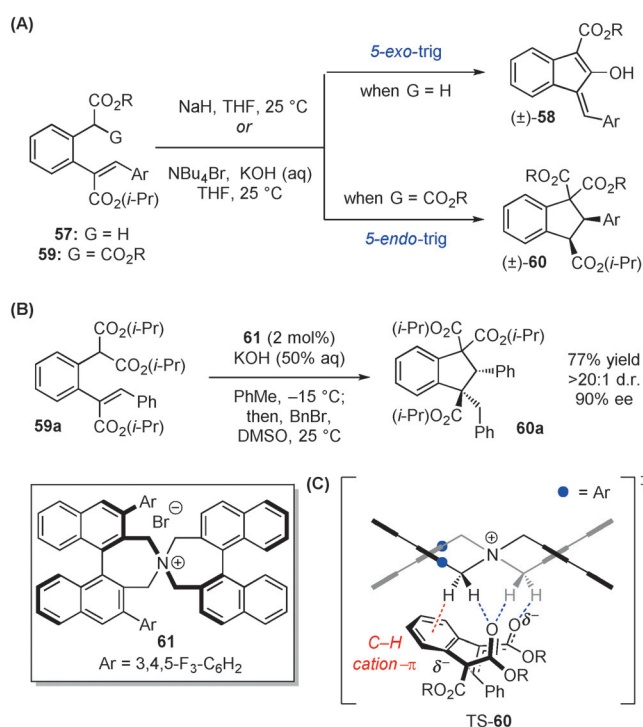
[b] Conversion of **54** after 24 h. [c] After 10 h.

a cation– $\pi$  interaction with the benzyl substituent and a Lewis acid–base interaction with the carbonyl oxygen atom (Scheme 16B). The importance of these attractive interactions was further corroborated by ground-state computational analysis, which indicated that tetrabutylammonium complexation favors the reactive substrate conformation leading to H-atom abstraction en route to formation of *trans*-dihydrobenzofuranol **55a**.

## 4.1.2.3. Chiral Ammonium Ion Catalyzed 5-endo-trig Cyclizations

Baldwin's rules for ring-closing reactions delineate the kinetic ease with which different cyclization modes occur

owing to the stereoelectronic constraints of bond formation between the reacting termini.<sup>[59]</sup> On the basis of these guidelines, the 5-*endo*-trig cyclization is generally considered to be highly unfavorable.<sup>[60]</sup> Paton, Smith, and co-workers recently reported unusual tetraalkylammonium-enabled 5-*endo*-trig intramolecular Michael reactions in the synthesis of substituted indanes **60**.<sup>[61]</sup> When simple  $\alpha$ -benzyl esters **57** are treated with base in the presence of achiral tetraalkylammonium ions, Dieckmann-type cyclization affords the traditional 5-*exo*-trig products (**58**). However,  $\alpha$ -benzyl malonates (**59**) subjected to the same conditions undergo exclusively 5-*endo*-trig cyclization (Scheme 17 A). The polysubstituted indane derivatives (**60**) generated through the action of chiral tetraalkylammonium catalyst **61** are correspondingly formed with high enantio- and diastereoselectivity (Scheme 17 B).



**Scheme 17.** A) 5-*exo*-trig vs. 5-*endo*-trig cyclization reactions. B) An intramolecular 5-*endo*-trig Michael reaction catalyzed by a chiral ammonium ion. C) The lowest-energy computed transition structure en route to **60** (B97D/6-31G(d)). The red dotted line represents a cation- $\pi$  interaction; blue dotted lines represent hydrogen bonds. Gray and black dotted lines indicate breaking or forming bonds.

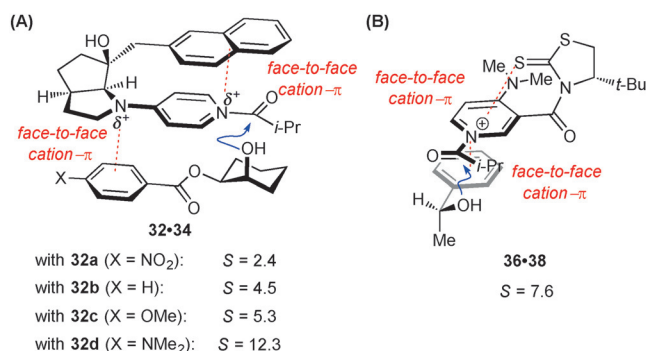
Computational studies into the origins of stereinduction revealed that the lowest-energy transition structure en route to the major product is characterized by three key interactions that define the arrangement of the catalyst-substrate tight ion pair (Scheme 17 C). The partially positive  $\alpha$ -C-H bonds of the ammonium ion form a cation- $\pi$  interaction with the substrate's aromatic linker and two non-classical hydrogen bonds to the carbonyl oxygen atoms of the malonate and Michael acceptor. These cooperative attractive interactions, which are largely absent in the minor transition structures, are proposed to underlie the transition-state stabilization responsible for

stereinduction. Consistent with this model, substrates with electron-withdrawing substituents on the aromatic linker undergo cyclization with diminished enantioselectivity.

#### 4.1.3. Heterocycle-Catalyzed Acyl Transfer Reactions

##### 4.1.3.1. Chiral Pyridine Catalyzed Kinetic Resolutions

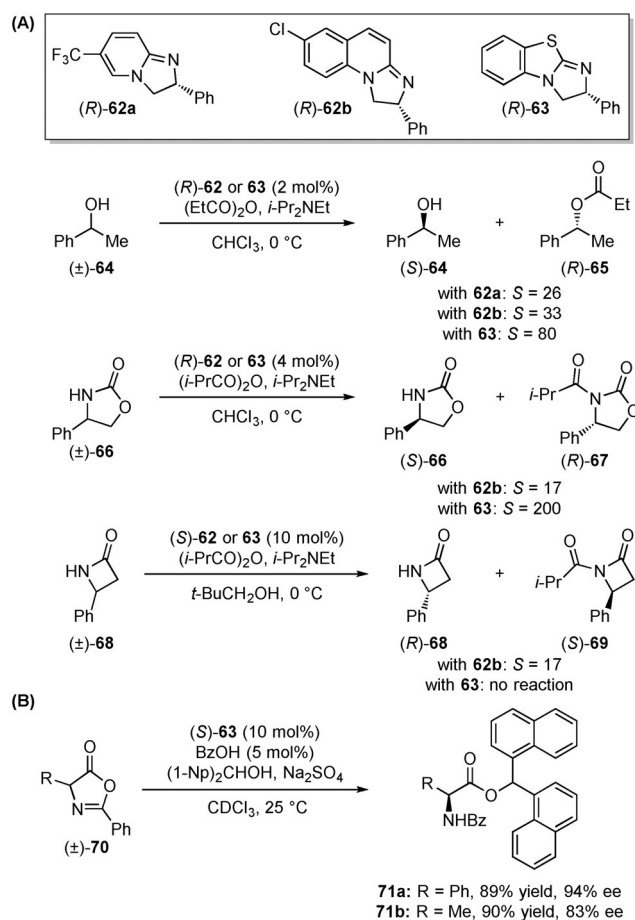
As discussed in Section 3.2.1, Fuji and co-workers disclosed naphthyl-substituted pyrrolidinopyridine catalyst **31** for the acylative kinetic resolution of benzoate-substituted secondary alcohols.<sup>[38]</sup> With substrates **32a–32d**, enantioselectivity tracks with the electron-donating ability of the benzoate substituent, such that alcohols bearing the most electron-rich group (**32d**) undergo acylation with a selectivity factor as high as  $S = 12.3$  (Figure 4 A). Discrimination between the enantiomers of the substrate was therefore attributed to an intermolecular cation- $\pi$  interaction between the benzoate directing group and the cationic acylated catalyst intermediate **34** on the face opposite the intramolecular cation- $\pi$  interaction controlling catalyst conformation. Yamada et al. proposed that a similar cation- $\pi$  interaction between the secondary benzylic alcohol substrate **36** and the acylated derivative of DMAP catalyst **38** organizes the reactive complex to enable discrimination between the substrate enantiomers with selectivity factors up to  $S = 30$  (Figure 4 B).<sup>[39]</sup>



**Figure 4.** Chiral 4-aminopyridine catalyst-substrate complexes en route to alcohol acylation. Red dotted lines indicate cation- $\pi$  interactions. Blue arrows indicate favorable trajectories for nucleophile approach.

##### 4.1.3.2. Kinetic Resolutions Catalyzed by Chiral Amidines and Isothioureas

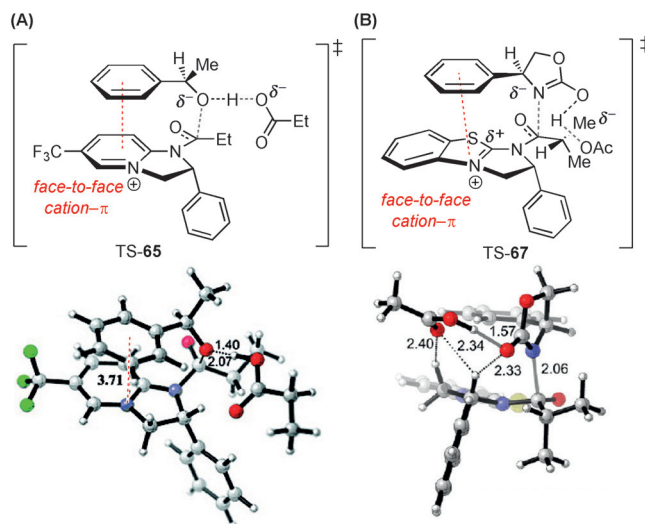
Inspired by the modularity and reactivity profile of imidazopyridine derivatives, Birman et al. crafted a family of structurally analogous chiral amidine (**62**) and isothiourea (**63**) catalysts for stereoselective acyl transfer reactions. The efficacy of these catalysts has been demonstrated in the acylative kinetic resolution of secondary alcohols,<sup>[62]</sup> oxazolidinones,<sup>[63]</sup> and lactams (Scheme 18 A).<sup>[64]</sup> These catalysts have also been employed for the dynamic kinetic resolution (DKR) of azlactones en route to natural and unnatural  $\alpha$ -amino esters (Scheme 18 B).<sup>[65]</sup>



**Scheme 18.** Amidine- and isothiourea-catalyzed acylative kinetic resolutions. Bz = benzoyl.

In each of these transformations, the  $\pi$ -donating functionality on the acyl acceptor plays a critical role in controlling reactivity and stereoselectivity, suggesting that cation– $\pi$  interactions with the acylated intermediate are important in the catalytic mechanism. To evaluate this hypothesis, computational (DFT) studies were directed towards assessing the basis for stereoselectivity in the kinetic resolution of alcohols and lactams (Figure 5).<sup>[64,66]</sup> In the lowest-energy transition structure located for the fast-reacting enantiomer of oxazolidinone **66**, the phenyl ring of the substrate is predicted to engage in a cation– $\pi$  interaction with the thiazolium moiety of isothiourea catalyst **63**, favoring reaction through a stacked conformation. This general arrangement was also found to be operative for reactions with amidine catalysts **62** across representative substrates, including 1-phenylethanol (**64**), 4-phenyloxazolidinone (**66**), and 4-phenylazetidinone (**68**).

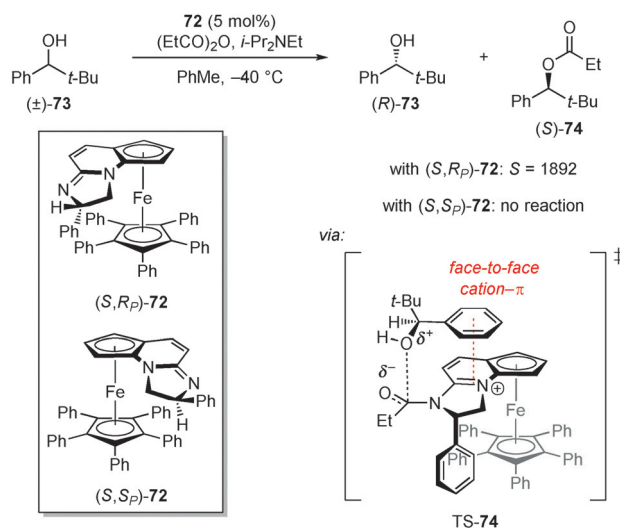
However, the basis for the absolute sense of the stereoselectivity observed in the DKR of azlactones required further investigation (Figure 6).<sup>[67]</sup> In the lowest-energy major and minor diastereomeric transition structures en route to **71a** and **71b**, strong cation– $\pi$  interactions between the benzhydryl group of the acyl acceptor and the acylated catalyst (ca. 3.3 Å) were found to organize the catalyst–substrate complexes. This organization enables C–C bond formation to be directed by additional stereodifferentiating



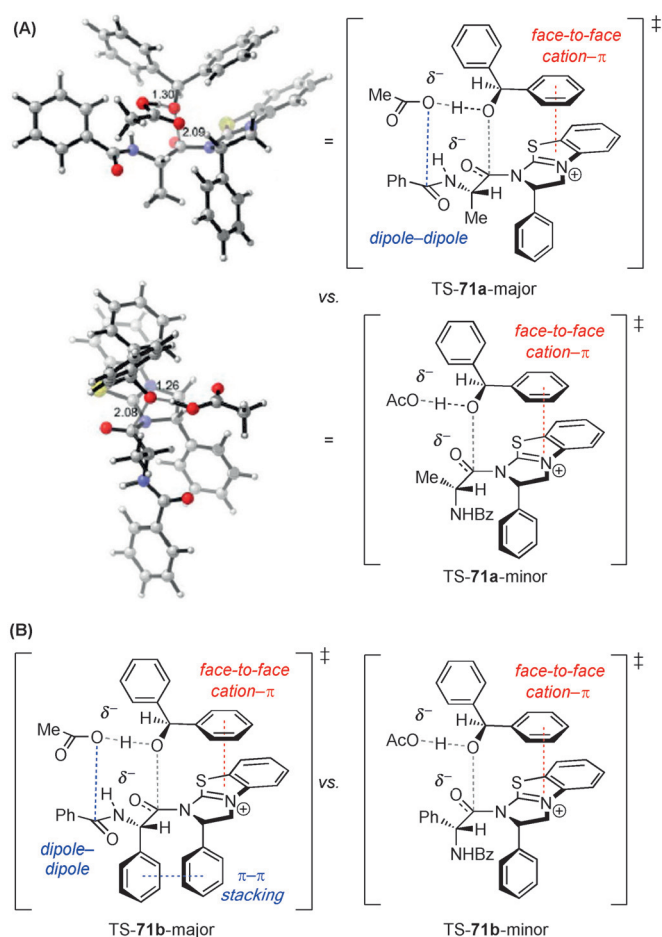
**Figure 5.** Lowest-energy computed transition structures for the acylation of the fast-reacting *R* enantiomers of A) alcohol **64** (B3LYP/6-31G(d)/CPCM(CHCl<sub>3</sub>)) and B) oxazolidinone **66** (M06-2X/6-31G(d)/SMD(CHCl<sub>3</sub>)). Red dotted lines represent cation– $\pi$  interactions. Black dotted lines represent other noncovalent interactions. Gray lines indicate breaking or forming bonds. Key distances shown in Å. C gray, H white, N blue, O red, F green, S yellow. Adapted with permission from Refs. [64] and [66]. Copyright 2008 and 2012 American Chemical Society.

interactions, including hydrogen bonding,  $\pi$ – $\pi$  stacking, and dipole–dipole interactions with a bridging carboxylate.

Fossey and co-workers subsequently combined insights from work by the groups of Fu<sup>[36]</sup> and Birman<sup>[62,66]</sup> to develop the ferrocene-based amidine catalyst **72**, which contains elements of planar and central chirality, for the acylative kinetic resolution of secondary benzylic alcohols (Scheme 19).<sup>[68]</sup> The stereoselectivity observed with this catalyst system is dependent on the matched arrangement of the pentaphenyl ferrocene floor and a second  $\alpha$ -aryl



**Scheme 19.** Acylative kinetic resolution of secondary benzylic alcohols by amidinium catalysts bearing matched planar- and central-chirality elements.



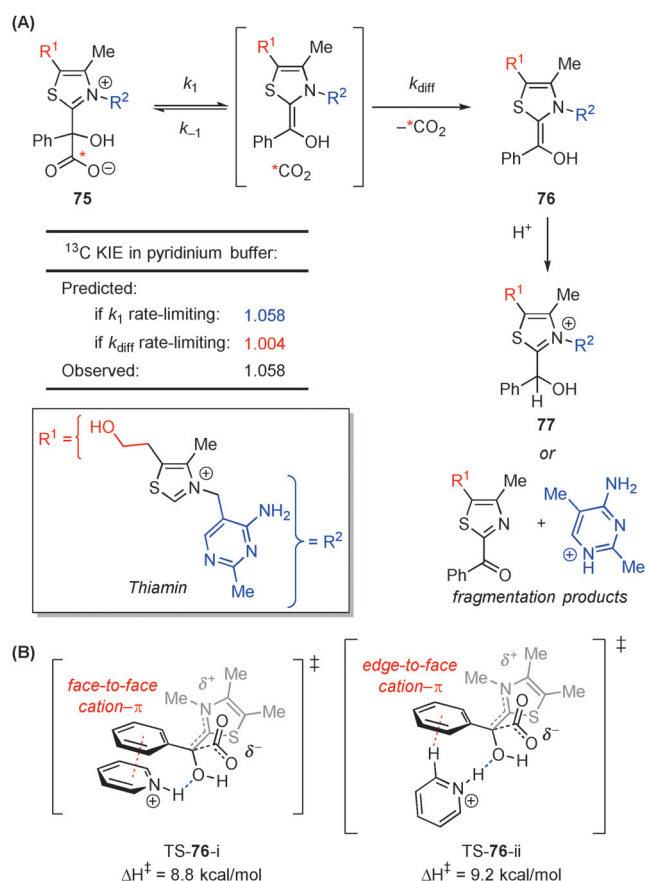
**Figure 6.** A) Lowest-energy computed transition structures for the acylation of the fast-reacting *S* enantiomer (top) and the slow-reacting *R* enantiomer (bottom) of substrate **70a** and B) substrate **70b** (M06-2X/6-311 + G(d,p)/SMD(CHCl<sub>3</sub>)/M06-2X/6-31G(d)/SMD(CHCl<sub>3</sub>)). Red dotted lines represent cation- $\pi$  interactions; blue dotted lines represent other noncovalent interactions. Gray and black dotted lines indicate breaking or forming bonds. Key distances shown in Å. C gray, H white, N blue, O red, S yellow. Adapted with permission from Ref. [67]. Copyright 2012 American Chemical Society.

appendage to shield the bottom face of the catalyst. The authors proposed that this arrangement forces acylation to proceed through an entirely stacked conformation. When the mismatched catalyst diastereomer is employed, the  $\alpha$ -aryl substituent of the catalyst precludes formation of this stacked arrangement, and no reaction occurs. These effects underscore the importance of the purported cation- $\pi$  interaction in substrate recognition, which is responsible for the impressive selectivity factors as high as  $S = 1892$ .

#### 4.1.4. Pyridinium Ion Catalyzed Decarboxylation

Mandelylthiamin (**75**) is a simple analogue of the first covalent intermediate formed by the thiamin diphosphate (TDP) dependent enzyme benzoylformate decarboxylase en route to the formation of benzaldehyde and CO<sub>2</sub>. While **75** undergoes decarboxylation in water at pH 5 to 7, the rate constant ( $k_{\text{cat}}$ ) is six orders of magnitude smaller than that of

the enzyme-catalyzed process. Furthermore, the immediate product of decarboxylation readily fragments with loss of the pyrimidine side chain. While a number of Brønsted acid additives reduce the degree of fragmentation, the conjugate acid of pyridine is uniquely effective at both accelerating decarboxylation and efficiently suppressing fragmentation (Scheme 20 A).<sup>[69]</sup>



**Scheme 20.** A) The agreement between predicted and experimental <sup>13</sup>C kinetic isotope effects indicates that C-C bond cleavage is the rate-limiting step of the decarboxylation mechanism. B) The two lowest-energy computed transition structures for the decarboxylation of mandelylthiamin in the presence of a pyridinium catalyst involve cation- $\pi$  interactions between the phenyl group and the pyridinium moiety. Red dotted lines represent cation- $\pi$  interactions; blue dotted lines represent hydrogen bonds. Predicted KIEs and enthalpies of activation calculated with M06-2X/6-31G\*\*/PCM(water).

In an effort to understand the pronounced impact of the simple pyridinium catalyst on the reaction profile, Singleton and Gonzales-James sought to elucidate the mechanistic features of this transformation.<sup>[70]</sup> Analysis of <sup>13</sup>C isotope effects revealed that, contrary to initial proposals, irreversible decarboxylation of **75** is rate-determining in both the catalyzed and uncatalyzed reaction mechanisms. Rapid CO<sub>2</sub> diffusion and subsequent protonation afford intermediates **76** and **77** en route to benzaldehyde and free thiamine.

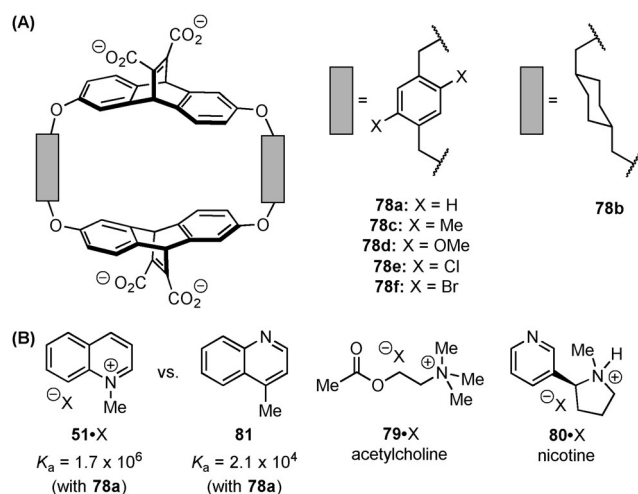
Computational (DFT and MP2) analysis indicated that both face-to-face and edge-to-face complexation of the

pyridinium catalyst and mandelylthiamin are highly favorable.<sup>[8]</sup> Furthermore, in cooperation with hydrogen-bond donation to the nascent enolate, this cation– $\pi$  interaction stabilizes the decarboxylation transition state, affording calculated rate accelerations in good agreement with experimental measurements (Scheme 20B). The computations also predict that this cooperative interaction is critical for effective transition-state stabilization, which is consistent with the experimental observation that no rate acceleration is observed with *N*-alkyl pyridinium ions or neutral Brønsted acids.

## 4.2. Catalyst $\pi$ -Donors

### 4.2.1. Cyclophane-Catalyzed Alkyl Transfer Reactions

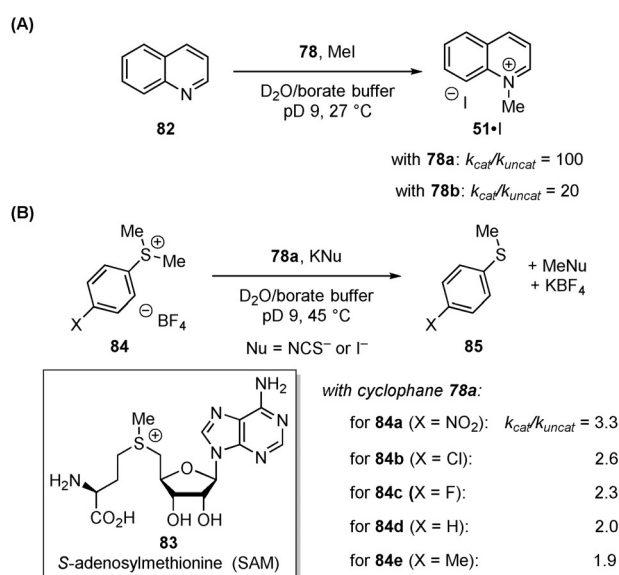
As part of their pioneering work investigating the thermodynamics of solution-phase cation– $\pi$  interactions, Dougherty and co-workers developed a set of water-soluble cyclophane-derived host molecules (**78**; Figure 7).<sup>[1,71]</sup> These



**Figure 7.** Representative A) cyclophane hosts and B) small-molecule guests.

hosts, which are characterized by their hydrophobic arene-lined cavities, recognize and desolvate cationic species, such as quinolinium (**51**) and ammonium ions (**79** and **80**), even in aqueous media.<sup>[5a,15a]</sup> Furthermore, the hosts exhibit a pronounced energetic preference for binding positively charged guests over their isostructural neutral analogues (i.e., **51** vs. **81**), indicating that the cation– $\pi$  interaction plays a central role in the recognition event.

Following the evaluation of the thermodynamics of guest complexation with cyclophane hosts **78**, Dougherty and co-workers sought to translate ground-state binding into transition-state stabilization, thereby utilizing cation– $\pi$  interactions to influence the reaction rate and/or selectivity. Towards this aim, they found that cyclophane **78a** imparts notable rate acceleration ( $k_{\text{cat}}/k_{\text{uncat}} = 100$ ) in the *N*-alkylation of quinoline (**82**) with methyl iodide in aqueous buffer (Scheme 21 A).<sup>[8a,72]</sup> The rate acceleration provided by the host could, in principle,



**Scheme 21.** Methyl transfer reactions catalyzed by cyclophanes **78**.

be explained simply on the basis of desolvation owing to the hydrophobic effect. However, cyclophane **78b**, which possesses cyclohexyldimethylene linkers rather than *para*-xylyl linkers, was found to exhibit a fivefold reduction in catalytic activity ( $k_{\text{cat}}/k_{\text{uncat}} = 20$ ). Given the importance of the  $\pi$ -donating ability of the host to catalysis, stabilizing cation– $\pi$  interactions between the  $\pi$ -faces lining the cyclophane and the positive charge developing in the alkylation transition state were also proposed to contribute to the rate acceleration. The transition-state binding energy was calculated from the rate acceleration ( $k_{\text{cat}}/k_{\text{uncat}}$ ) to enable comparison with the binding affinity for *N*-methyl quinolinium ions (**51**·X). Remarkably, the authors observed that the cyclophane hosts exhibit a greater binding affinity for the transition state, which bears only a partial positive charge, than for the product, which bears a full positive charge.

Dougherty and co-workers extended the application of these cyclophane catalysts to biomimetic methyl-transfer reactions with sulfonium ion electrophiles resembling the ubiquitous cofactor S-adenosylmethionine (SAM, **83**), which commonly engages in cation– $\pi$  interactions in biological systems (Scheme 21 B).<sup>[8,13,73]</sup> Demethylation of sulfonium **84** by an anionic nucleophile to afford sulfide **85** involves a net decrease in formal charge. Therefore, it might be expected that a catalyst designed to stabilize positive charge would be ineffective in this reaction, as it might bind the ground-state starting material more tightly than the rate-limiting transition structure. However, cyclophane **78a** was found to accelerate the reaction by a factor of three. To rationalize this observation, the authors proposed that an interplay of electrostatic<sup>[7]</sup> and polarizability effects governs the overall strength of the cation– $\pi$  interactions.<sup>[8]</sup> As transition structures generally have longer bonds and more diffuse charge distributions than their substrate and product ground states, they are effectively more polarizable. This increased transition-state polarizability was proposed to compensate for the decreased cationic character in the sulfonium ion demethy-

lation, thereby enabling the cyclophane host to promote the reaction by binding the transition state more tightly than the precursor ground state, effectively lowering the energy of activation.

The validity of this hypothesis was evaluated by a Hammett analysis of the demethylation rate of substituted sulfonium ions (**84a–84e**), both in the presence and absence of cyclophane host **78a**. While both  $k_{\text{cat}}$  and  $k_{\text{uncat}}$  increase with the incorporation of electron-withdrawing substituents (increasing  $\sigma$ ) on the sulfonium ion, the slope ( $\rho$ ) is greater in the presence of the cyclophane host. As such, the host enhances the substituent effect on the reaction rate; the greatest acceleration was observed with electron-deficient substrates, which react through early, highly cationic transition states.<sup>[8a]</sup> Furthermore, substituted cyclophanes (**78a**, **78c–78f**) spanning a range of polarizabilities show marked substituent effects in the dealkylation of **84a** (Table 4).<sup>[8b]</sup>

**Table 4:** Correlation between polarizability, binding, and catalysis of sulfonium ion demethylation.

| Entry | Host       | Polarizability <sup>[a]</sup> | $-\Delta G_a$ <sup>[b]</sup><br>[kcal mol <sup>-1</sup> ] | $k_{\text{cat}}/k_{\text{uncat}}$ | $-\Delta\Delta G^\ddagger$ <sup>[c]</sup><br>[kcal mol <sup>-1</sup> ] |
|-------|------------|-------------------------------|-----------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------|
| 1     | <b>78a</b> | 10.3 (PhH)                    | 5.7                                                       | 4.9                               | 1.0                                                                    |
| 2     | <b>78c</b> | 12.3 (PhMe)                   | 5.2                                                       | 6.4                               | 1.2                                                                    |
| 3     | <b>78d</b> | 13.1 (PhOMe)                  | 4.6                                                       | 9.7                               | 1.4                                                                    |
| 4     | <b>78e</b> | 14.1 (PhCl)                   | 5.8                                                       | 9.8                               | 1.4                                                                    |
| 5     | <b>78f</b> | 14.7 (PhBr)                   | 5.6                                                       | 12.0                              | 1.6                                                                    |

[a] Polarizability of the corresponding arene with the same substituent (indicated in parentheses). [b]  $\Delta G_a$ : the free energy of association at 298 K. [c]  $\Delta\Delta G^\ddagger$ : the differential free energy of activation between the catalyzed and uncatalyzed pathways at 319 K. See Scheme 21 B.

Whereas hosts **78a** and **78c–78f** show little difference in ground-state binding affinity ( $\Delta G_a$ ), the substituent-dependent rate acceleration ( $k_{\text{cat}}/k_{\text{uncat}}$ ) correlates well with the polarizability of the substituted linker (Table 4). Taken together, these results provide compelling evidence that cation– $\pi$  interactions contribute substantially to catalysis, wherein differential stabilization of the transition state over the ground state depends both on electrostatic attraction and on the polarizability of the  $\pi$ -donor and  $\pi$ -acceptor.

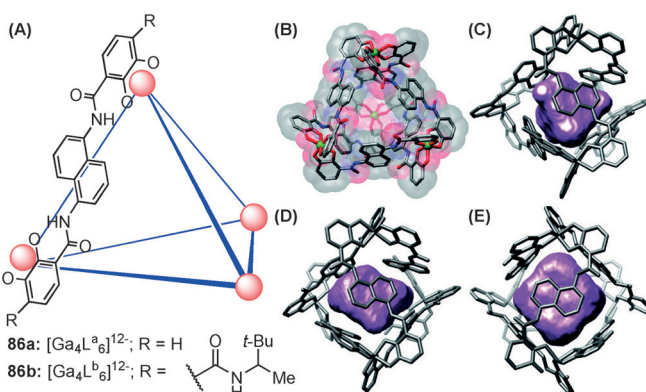
#### 4.2.2. Reactions Mediated by Supramolecular Assemblies

The design and development of hosts that behave as molecular vessels for reacting guest partners is a longstanding goal in the field of supramolecular chemistry. Utilizing the hydrophobic effect to bind small-molecule guests, these supramolecular hosts can create unique chemical microenvironments, distinct from that of the bulk solution, which enable reactions to take place with accelerated rates or altered selectivity. The rate and selectivity enhancements in such systems are inherently linked to the nature of the constrained binding cavity, which imparts strict size discrimination while potentially increasing the effective molarity of reactants, thereby significantly reducing the entropic cost of the reaction. Furthermore, cation– $\pi$  interactions involving the aromatic groups lining the binding cavities of many hosts play

a crucial role in determining the strength, orientation, and selectivity of guest-binding events. Elegant demonstrations of the catalytic potential of such supramolecular assemblies abound.<sup>[74]</sup>

##### 4.2.2.1. Cation– $\pi$ Interactions in Guest Encapsulation

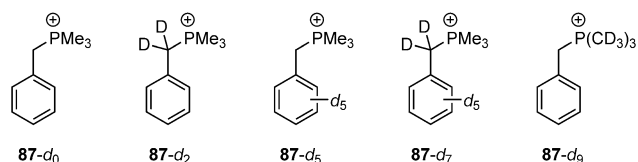
As part of a broader program on developing supramolecular assemblies for exploring host–guest chemistry, Raymond and co-workers designed a series of tetrahedral molecular vessels ( $M_4L_6$ , **86**) that self-assemble upon combination of trivalent metal ions ( $M = \text{Ga}^{3+}$ ,  $\text{Al}^{3+}$ ,  $\text{In}^{3+}$ , or  $\text{Fe}^{3+}$ ) and naphthalene-bridged bis(bidentate) catechol ligands (Figure 8A).<sup>[75]</sup> These dodecaanionic supramolecular scaffolds



**Figure 8.** A) Schematic representation of the  $[M_4L_6]^{12-}$  assembly, with one molecule of ligand  $L^a$  shown explicitly. B) A space-filling model of **86a**. Crystal structures showing C)  $\text{NEt}_4^+$ , D)  $\text{BnNMe}_3^+$ , and E)  $[\text{Cp}^*_2\text{Co}]^+$  ions encapsulated in  $\text{Ga}_4\text{L}_6$  (**86a**).  $\text{Cp}^* = \text{pentamethylcyclopentadienyl}$ . Adapted with permission from Ref. [77]. Copyright 2009 American Chemical Society.

are soluble in water, yet they contain a central arene-lined hydrophobic cavity spanning  $0.35\text{--}0.5\text{ nm}^3$  (Figure 8B). As such, these supramolecular clusters bind a range of positively charged organic and organometallic compounds with high affinity (Figure 8C–E). For example, appropriately sized phosphonium ions, ammonium ions, and iminium ions are encapsulated with association constants as high as  $10^5\text{ M}^{-1}$  in aqueous media.<sup>[76]</sup> Furthermore, crystallographic analysis of a cluster-encapsulated benzyltrimethylammonium ion ( $\text{BnNMe}_3^+\text{C}86\text{a}$ ) revealed a close contact ( $3.63\text{ \AA}$ ) between the cation and the naphthalene walls of the cluster, suggesting that cation– $\pi$  interactions likely contribute to the overall binding strength (Figure 8D).<sup>[77]</sup>

The role of the cation– $\pi$  interaction in guest binding was probed quantitatively through equilibrium isotope effect (EIE) studies using a series of benzyltrimethylphosphonium isotopologues (**87-d<sub>n</sub>**; Figure 9).<sup>[78]</sup> Host **86a** exhibits a greater affinity for the protiated isotopologue **87-d<sub>0</sub>** than for its deuterated isotopologues (i.e.,  $K_{d_0}/K_{d_n} \geq 1$ ; Table 5), and Van't Hoff analysis indicates that the differential binding affinity is enthalpically determined. These EIE values arise from changes in the guest vibrational force constants and zero-point energies, which result from attractive interactions



**Figure 9.** Isotopologues of the benzyltrimethylphosphonium ion.

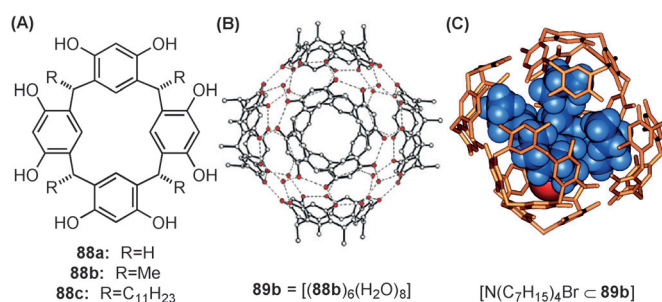
**Table 5:** EIE values for the interior binding of guest **87-d<sub>n</sub>** to host **86a**.

| Entry | Isotopologue <sup>[a]</sup> | EIE ( $K_{d0}/K_{dn}$ ) <sup>[b]</sup> | EIE per D <sup>[c]</sup> |
|-------|-----------------------------|----------------------------------------|--------------------------|
| 1     | <b>87-d<sub>2</sub></b>     | 1.07(1)                                | 1.034(6)                 |
| 2     | <b>87-d<sub>5</sub></b>     | 1.00(2)                                | 1.000(3)                 |
| 3     | <b>87-d<sub>7</sub></b>     | 1.103(7)                               | 1.014(1)                 |
| 4     | <b>87-d<sub>9</sub></b>     | 1.14(1)                                | 1.015(1)                 |

[a] See Figure 9. [b] Measured in D<sub>2</sub>O at 298 K. [c] EIE per D =  $(K_{d0}/K_{dn})^{1/n}$ .

weakening these C–H/D bonds upon binding to the host. Furthermore, larger EIEs per deuterium are observed at the methyl and benzylic C–H/D positions compared to the aromatic positions. Given that the methyl and benzylic C–H/D bonds bear a significant portion of the net charge of the phosphonium ion, while the aromatic C–H/D bonds do not, these EIEs are consistent with specific, enthalpically stabilizing cation– $\pi$  interactions between these bonds and the arenes lining the host cavity. Theoretical EIE values calculated using DFT are in good agreement with the experimental values and further validate this interpretation.

In an independent effort, MacGillivray and Atwood introduced the spherical molecular vessel **89a**, which comprises six calix[4]resorcinarene subunits (**88**) that self-assemble through hydrogen-bonding interactions in water-saturated, nonpolar media (Figure 10A).<sup>[79]</sup> This neutral supramolecular scaffold and its alkyl-substituted derivatives (**89b**, **89c**) form hydrophobic arene-lined cavities with volumes of approximately 1.4 nm<sup>3</sup> (Figure 10B), and persist both in the solid state and in water-saturated chloroform.<sup>[80]</sup> Rebek and co-workers subsequently demonstrated that **89b** and **89c** bind cationic guests, such as tetraalkylammonium ions, with high



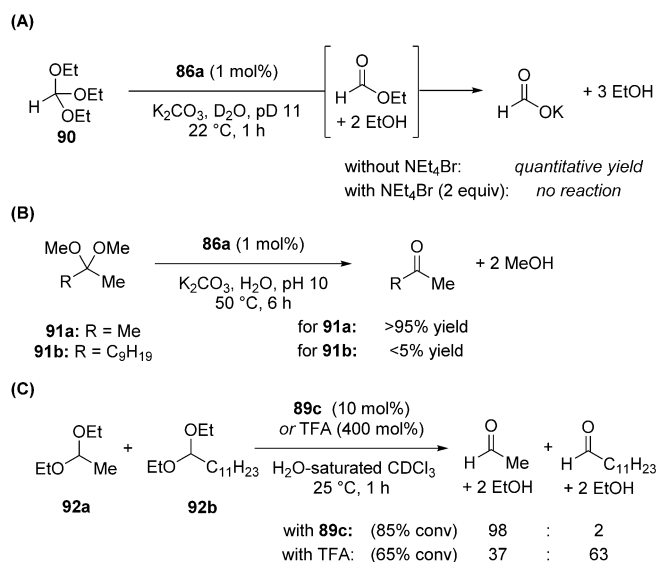
**Figure 10.** A) One calix[4]resorcinarene subunit. B) Crystal structure of **89b**, which comprises six calix[4]resorcinarene subunits and eight water molecules. Adapted with permission from Ref. [80c]. Copyright 2004 American Chemical Society. C) Energy-minimized structure of tetraheptylammonium bromide encapsulated within **88b**. Methyl substituents omitted for clarity. Image adapted from Ref. [80a].

affinity (Figure 10C). They attributed this favorable cation encapsulation to the combined effect of enthalpically stabilizing cation– $\pi$  interactions between the guests and the aromatic walls of the capsule and the entropically stabilizing release of encapsulated solvent molecules.<sup>[80b,c]</sup>

Subsequent studies revealed that the phenolic residues of **89c** act as Brønsted acids ( $pK_a$  5.5–6) to bind neutral, basic guests as their cationic conjugate acids.<sup>[81]</sup> For example, triethylamine is encapsulated within the conjugate base of **89c** as the dimeric ammonium ion (i.e., [Et<sub>3</sub>N·HNET<sub>3</sub>]<sup>+</sup>). Tiefenbacher and Zhang attributed this increase in acidity relative to free resorcinol ( $pK_a \approx 9$ ) to the energetic preference for the encapsulation of cationic guests owing to the formation of stabilizing cation– $\pi$  interactions with the arenes lining the central cavity. Given their demonstrated ability to stabilize cationic species, assemblies **86** and **89** have been the focus of growing interest for the catalysis of reactions involving cationic substrates and/or intermediates.

#### 4.2.2.2. Acid Catalysis Mediated by Supramolecular Vessels

Whereas both orthoformates and acetals are generally stable in neutral and basic media, they undergo rapid hydrolysis in acidic solution. Inspired by the observation that **86a** can increase the basicity of encapsulated guests by up to 4.5  $pK_a$  units, Pluth et al. employed the complex as a catalyst for the hydrolysis of orthoformates and acetals (Scheme 22A,B).<sup>[82]</sup> In the presence of **86a**, triethyl orthoformate (**90**) and dimethoxypropane (**91a**) rapidly undergo hydrolysis, even under basic conditions, to afford the corresponding formate ester, which is readily hydrolyzed outside the vessel to afford formate and acetone, respectively. These reactions are inhibited by tetrabutylammonium bromide, which is known to block the cavity of vessel **86a**, and the substrate scope displays a pronounced size dependence wherein only sufficiently compact substrates undergo hydroly-



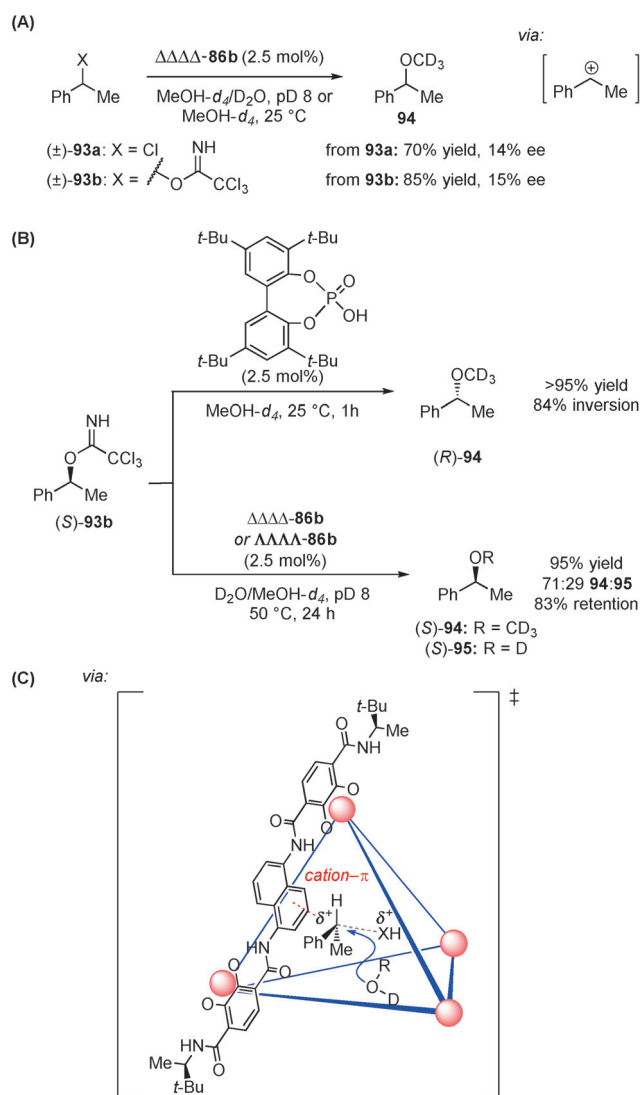
**Scheme 22.** Hydrolysis of acid-labile functional groups mediated by supramolecular vessels.

ysis. These observations are consistent with the proposal that the reaction occurs exclusively within the inner cavity of the host.

To elucidate the mechanism of catalysis, these reactions were monitored by  $^1\text{H}$  NMR spectroscopy. This analysis revealed that the resonances of the naphthalene protons on the bridging ligand of **86a** undergo a downfield shift in the presence of reactive substrates whereas no change occurs in the presence of unreactive orthoformates.<sup>[82b]</sup> This downfield shift provides evidence for a specific substrate–vessel interaction, which was determined to be the complex with neutral substrate (**89** or **90**–**86a**). Whereas the corresponding cationic protonated intermediates have not been observed spectroscopically, the authors propose that the stabilizing interaction would be strengthened upon the development of positive charge, thereby contributing to acceleration of the rate-limiting protonation (as in orthoformate hydrolysis) or nucleophile addition (as in acetal hydrolysis).

The resorcinarene-derived capsule **89c** was also demonstrated to be a competent catalyst for the hydrolysis of acetals<sup>[81]</sup> and the hydration of isonitriles<sup>[83]</sup> in water-saturated chloroform (Scheme 22C). As observed with **86a**, the reactions catalyzed by **89c** were inhibited by exogenous tetrabutylammonium bromide and exhibited pronounced size selectivity. The differential rates of substrate ingress and egress were demonstrated to enable the selective hydrolysis of one acetal in the presence of another. The selective yet efficient reactivity observed under mild conditions was attributed to Brønsted acid catalysis by the phenol residues of the capsule and by stabilizing cation– $\pi$  interactions with the resultant cationic intermediates and transition structures.

In an extension of the Brønsted acid catalysis enabled by these supramolecular assemblies, Zhao et al. demonstrated that tetrahedral assembly **86b** catalyzes the solvolysis of secondary benzylic chlorides (**93a**) and trichloroacetimidates (**93b**).<sup>[84]</sup> The formation of enantioenriched product was achieved from racemic **93**, albeit with very low levels of stereoinduction, in the presence of a single enantiomer of the capsule (either  $\Delta\Delta\Delta\Delta$ -**86b** or  $\Lambda\Lambda\Lambda\Lambda$ -**86b**; Scheme 23A). As the identity of the leaving group (trichloroacetimidate vs. chloride) was observed to be inconsequential to the degree of enantioenrichment, the authors proposed that both substrates react through a shared intermediate, presumed to be the secondary benzylic carbocation. However, they also note that the reaction of a single enantiomer of substrate **93a** or **93b** in the presence of either enantiomer of **86b** yields the solvolysis products **94** and **95** with predominantly retention of stereochemistry (Scheme 23B). Whereas secondary benzylic electrophiles typically undergo  $\text{S}_{\text{N}}2$  solvolysis with stereoinversion, the authors advanced that an  $\text{S}_{\text{N}}1$  mechanism could account for the unusual stereochemical outcome. They proposed a transition structure in which cation– $\pi$  interactions between the naphthalene walls of the cavity and the positive charge developing at the benzylic carbon atom reduce the barrier to ionization (Scheme 23C). Preservation of this interaction with the resultant carbocationic intermediate would shield one of its faces, forcing subsequent rapid nucleophile addition to occur from the same face as leaving-group departure to afford the product with net stereoreten-

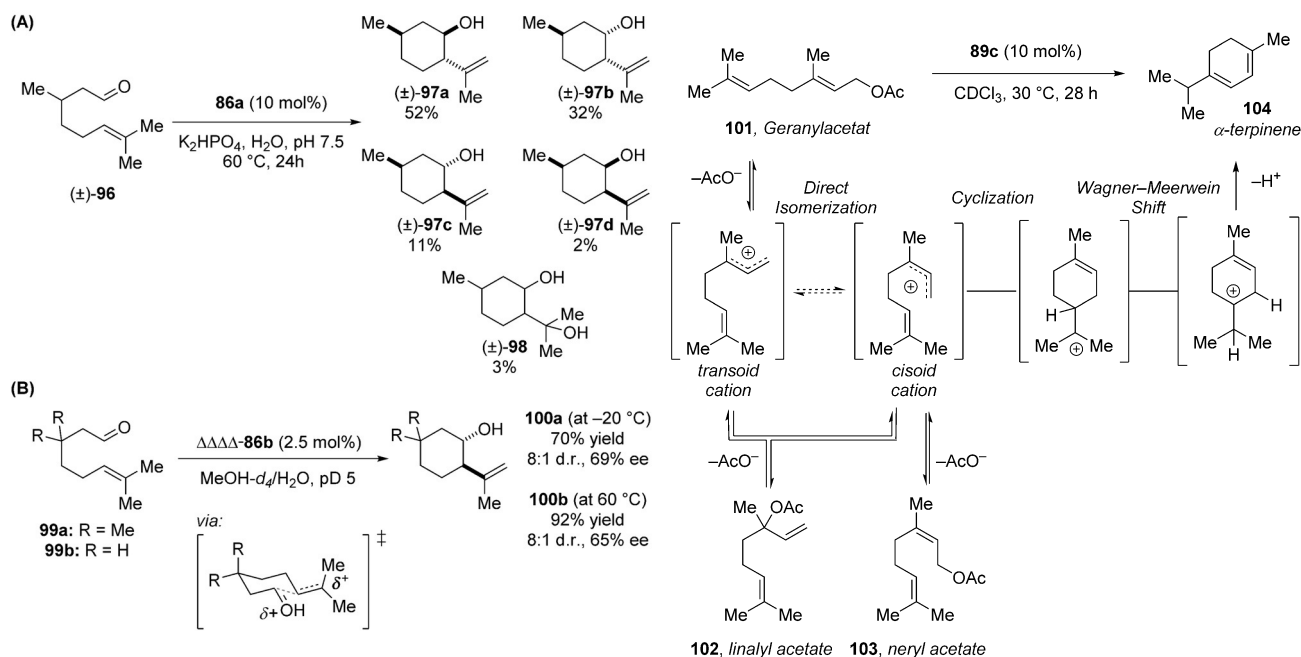


**Scheme 23.** Stereoretentive solvolysis of secondary benzylic trichloroacetimidates and chlorides catalyzed by supramolecular vessels. The red dotted line represents a cation– $\pi$  interaction; the gray dotted line represents the breaking bond. The blue arrow represents the trajectory accessible to the nucleophile in the subsequent bond-forming step.

tion. This work provides a compelling example of the impact that stabilizing cation– $\pi$  interactions can exert on reaction outcomes.

#### 4.2.2.3. Cyclization Reactions Catalyzed by Supramolecular Vessels

Inspired by the head-to-tail terpene cyclase enzymes, which exercise exquisite control over C–C bond formation from cationic intermediates through networks of cation– $\pi$  and other noncovalent interactions, Hart-Cooper et al. sought to apply Brønsted acid catalysis enabled by **86** to the Prins cyclization of citronellal and related substrates.<sup>[85]</sup> Assembly **86a** was observed to catalyze the Prins cyclization of citronellal (**96**) to afford isopulegol (**97a**) and its stereoisomers (**97b**–**97d**; Scheme 24A). Unlike the acid-catalyzed reaction in the absence of **86**, the reaction proceeding within



**Scheme 24.** Prins cyclization reactions catalyzed by supramolecular vessels.

the hydrophobic cavity of the catalyst affords the cyclic elimination products (**97**) rather than their hydrated analogues (**98**). Furthermore, using enantiopure assembly **86b** enables enantioselective Prins cyclizations of achiral citronellal analogues (**99a**, **99b**) in up to 69% *ee* (Scheme 24 B).<sup>[85b,c]</sup> Enantioinduction is proposed to be due to interactions between the cationic C–C bond-forming transition structure and the naphthalene spacer of the chiral assembly.

Tiefenbacher and Zhang were similarly inspired by the tail-to-head terpene cyclase enzymes to apply hexameric resorcinarene capsule **89c** to the cyclization of monoterpenes, including geraniol, nerol, and linalool, and their corresponding acetates (**101–103**).<sup>[86]</sup> Whereas tail-to-head terpene cyclization is common for enzyme-catalyzed reactions, this mode of reactivity has been challenging to reproduce with synthetic systems. Remarkably, the authors were able to demonstrate that assembly **89c** effectively catalyzes the “non-stop” cyclization of geranyl acetate (**101**) to selectively afford  $\alpha$ -terpinene (**104**; Scheme 25). To account for the long lifetime of the cationic  $\alpha$ -terpinyl intermediate, which is sufficient to allow for a 1,2-Wagner–Meerwein shift, the authors invoked stabilizing cation– $\pi$  interactions between the intermediate and the aromatic walls of the capsule. They also proposed that analogous stabilizing cation– $\pi$  interactions in the transition structure are responsible for reducing the activation barrier for the direct isomerization of the transoid allylic cation intermediate into the cisoid form, allowing for cyclization directly from geranyl acetate and bypassing the linalyl acetate intermediate previously considered to be necessary. The disparate product distributions and reaction rates observed with geranyl, neryl, and linalyl acetate under identical reaction conditions are consistent with this proposal. This mechanism of catalysis is loosely analogous to that

**Scheme 25.** Tail-to-head terpene cyclization catalyzed by a supramolecular vessel.

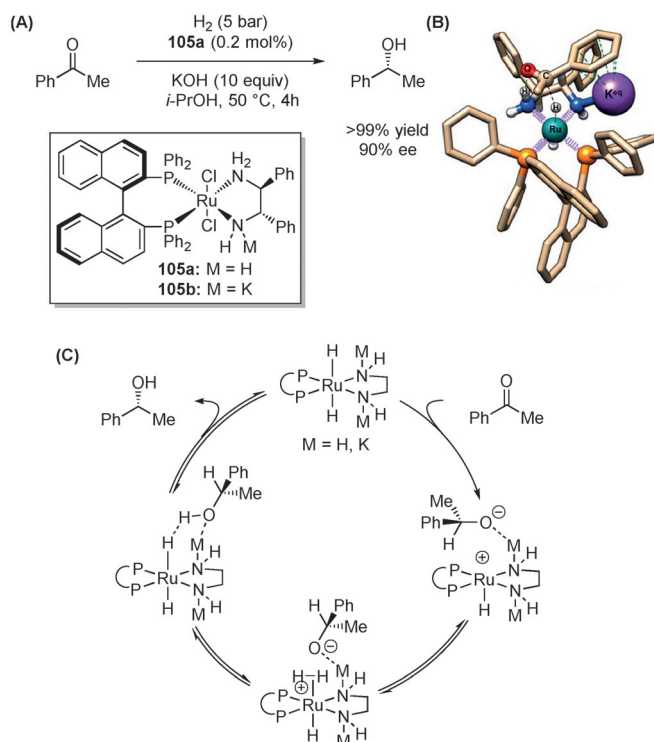
employed by terpene cyclase enzymes, which control reaction outcomes by selectively stabilizing cationic intermediates and transition structures through multiple interactions with aromatic residues lining the active site.<sup>[87]</sup>

#### 4.2.2.4. Transformations Catalyzed by Encapsulated Transition-Metal Complexes

In a collaborative effort, the Raymond, Bergman, and Toste groups have also explored the use of supramolecular assemblies **86a** and **86b** to bind cationic organometallic complexes that are capable of catalyzing a range of transformations.<sup>[74g,88]</sup> Although the precise nature of the interactions between these cationic organometallic complexes and the cavity of the supramolecular host requires further elucidation, the potential of these encapsulated catalyst systems to enable both enthalpic and entropic transition-state stabilization offers tantalizing opportunities for future development.

#### 4.2.3. Alkali Metal Cation Assisted Ruthenium-Catalyzed Asymmetric Hydrogenation Reactions

Since Noyori's seminal report,<sup>[89]</sup> asymmetric ruthenium-catalyzed ketone hydrogenation has become a powerful and practical method for the preparation of highly enantioenriched chiral secondary alcohols (Scheme 26 A). These reactions typically require superstoichiometric quantities of an alkali metal hydroxide or alkoxide additive to attain optimal rates. Initial proposals suggested that the alkoxide may serve to neutralize the HCl formed during the activation of the ruthenium precatalyst or to accelerate deprotonation of



**Scheme 26.** A) Ruthenium-catalyzed hydrogenation of acetophenone. B) The lowest-energy computed transition structure ( $\omega$ B97X-D/SDD-(Ru)/6-31G\*(C,H,N,O,P,K)/SMD(propan-2-ol)) for rate- and selectivity-determining C–H bond formation catalyzed by amidato complex **105b**. The proposed cation– $\pi$  interaction is depicted by green dotted lines. Adapted with permission from Ref. [91]. Copyright 2014 American Chemical Society. C) Revised catalytic cycle.

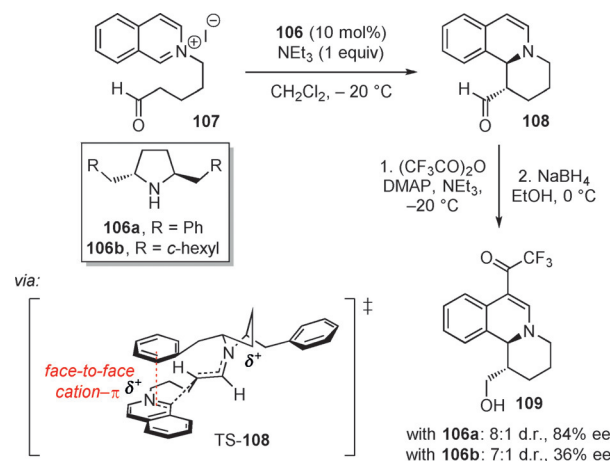
a purported  $\eta^2$ -H<sub>2</sub> Ru complex. However, Hartmann and Chen demonstrated that the Lewis acidic alkali metal cation is also essential for rapid turnover.<sup>[90]</sup> Nevertheless, its precise role in the catalytic cycle remained ambiguous.

In recent work to elucidate the role of the metal alkoxide, Dub et al. proposed a modification to the conventionally accepted mechanism, thereby reconciling it with several prior experimental studies and their own computational results.<sup>[91]</sup> According to a reaction coordinate analysis for the enantioselective reduction of acetophenone with *trans*-[RuH<sub>2</sub>][(S,S)-BINAP][(S,S)-DPEN]] (**105a**; BINAP = 2,2'-bis(diphenylphosphino)-1,1'-binaphthalene, DPEN = 1,2-diphenyl-1,2-diaminoethane), outer-sphere hydride transfer was calculated to be the rate- and enantiodetermining step. The rate acceleration induced by a large excess of potassium *tert*-butoxide was attributed to the equilibrium formation of potassium amidato Ru complex **105b**, which is stabilized by cation– $\pi$  interactions between the alkali metal counteranion and the aryl groups on both the DPEN and BINAP ligands (Scheme 26B). Reaction through this potassium amidato complex reduces the barriers to rate-determining hydride transfer by 1–2 kcal mol<sup>–1</sup> and to subsequent H–H bond cleavage by 10 kcal mol<sup>–1</sup> (Scheme 26C). Consistent with the importance of the cation– $\pi$  interaction, the experimentally determined catalytic activity was found to depend on the identity of the cation ( $K^+ > Na^+ \approx Rb^+ > Li^+$ ),<sup>[90]</sup> in striking

agreement with the relative strengths of the metal cation– $\pi$  interactions in aqueous solution ( $K^+ > Rb^+ > Na^+ > Li^+$ ).<sup>[11a]</sup>

#### 4.2.4. Pyrrolidine-Catalyzed Mannich Annulation Reactions

Jørgensen and co-workers have developed a Mannich-type annulation strategy for the enantio- and diastereoselective construction of 1,2-dihydroisoquinolines (**108**; Scheme 27).<sup>[92]</sup> The C<sub>2</sub>-symmetric secondary amine catalyst

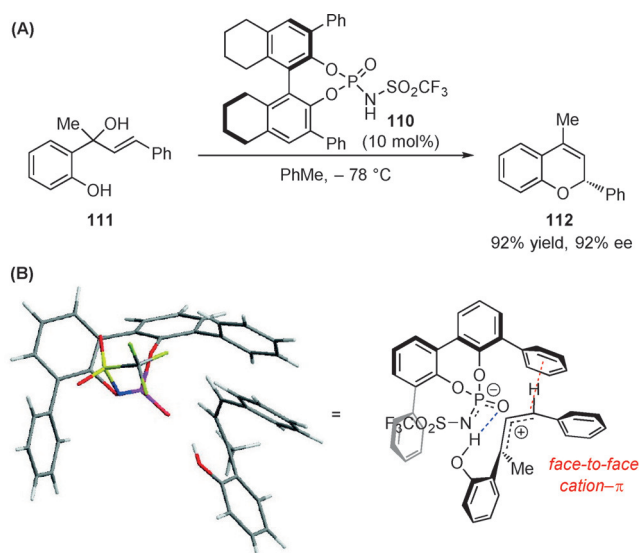


**Scheme 27.** Chiral pyrrolidine catalyzed asymmetric Mannich reaction with a stereochemical model for the transition structure leading to the major product. The red dotted line represents a cation– $\pi$  interaction. Black dotted lines represent breaking or forming bonds. DMAP = 4-dimethylaminopyridine.

**106a** activates 2-(5-oxopentyl)isoquinolinium derivatives such as **107** through enamine formation, thereby triggering cyclization onto the *N*-alkyl isoquinolinium core. In this process, the pendent benzyl groups on the amine catalyst are essential for good stereoselection. Catalyst **106b**, which bears cyclohexylmethylene rather than benzyl motifs, affords **109** with similar diastereoselectivity but markedly reduced enantioselectivity. To explain this observation, Jørgensen and co-workers proposed a stereochemical model in which an attractive cation– $\pi$  interaction between the benzyl group of the catalyst and the isoquinolinium cation favors reaction through a stacked conformation. This conformation selectively exposes the *Si* face of the isoquinolinium ion, and poises the enamine to undergo addition to form (*S,S*)-**108**, in accordance with the experimentally determined absolute configuration.

#### 4.2.5. Phosphoramidate-Catalyzed Allylic Etherifications

Achieving high enantioinduction in reactions proceeding through unstabilized carbocationic intermediates presents a major challenge in selective catalysis. Kinetic differentiation of diastereomeric reaction manifolds involving these highly reactive electrophiles is difficult owing to the very early nature of the transition states en route to stable products. In a step towards harnessing the reactivity of such carbocations, Rueping and co-workers employed the chiral phosphoramidate



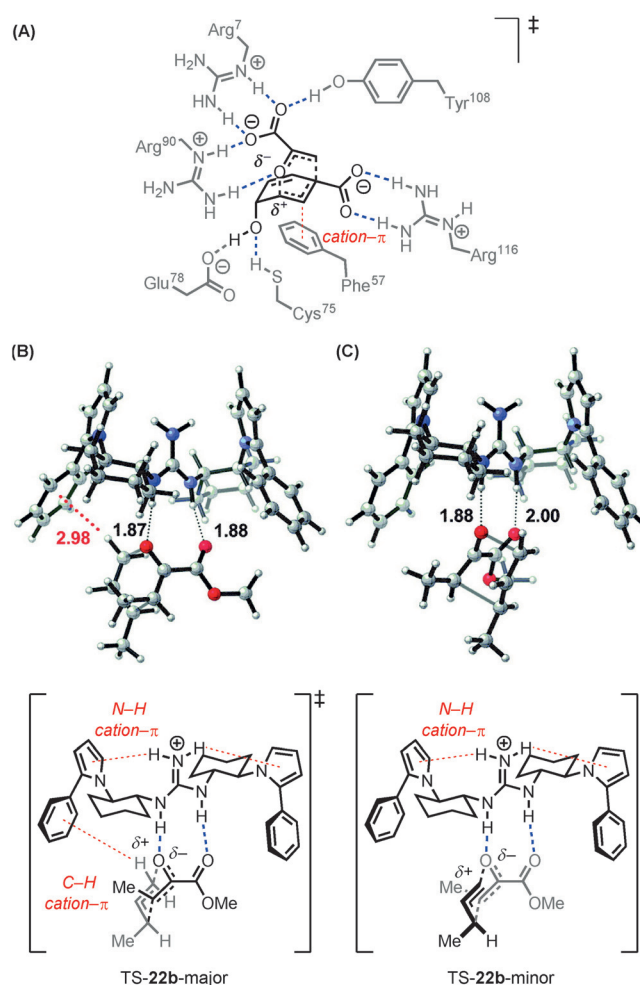
**Scheme 28.** A) Phosphoramidate-catalyzed etherification of allylic alcohols with B) a stereochemical model of the carbocationic intermediate en route to major product **112**. The red dotted line represents a cation- $\pi$  interaction. The blue dotted line represents a hydrogen bond. The cyclohexyl rings of catalyst **110** are omitted for clarity. C gray, H white, N blue, O red, P purple, S yellow. Adapted with permission from Ref. [93]. Copyright 2011 American Chemical Society.

Brønsted acid catalyst **110** for  $S_N1'$  substitution reactions of tertiary benzylic cinnamyl alcohols (e.g., **111**) to form enantioenriched 2-arylchromenes (**112**; Scheme 28).<sup>[93]</sup>

In this example of asymmetric ion-pair catalysis, the chiral acid is proposed to protonate **111**, triggering dehydration and the formation of an allylic cation. This electrophile is stabilized through electrostatic interactions with the chiral phosphoramidate counteranion, which then directs cyclization to afford **112** with good enantiomeric excess. Notably, asymmetric induction is highly dependent on the substituents at the 3,3'-positions of the BINOL (1,1'-binaphthalene-2,2'-diol) framework. Only catalysts with phenyl groups at these sites catalyze the cyclization with good enantiocontrol. The authors postulated that this effect is due to an enantiodetermining cation- $\pi$  interaction between these phenyl substituents and the allylic cation, orienting the substrate appropriately for highly enantioselective addition of the phenol nucleophile.

#### 4.2.6. Guanidinium Ion Catalyzed Claisen Rearrangements

In the first known step of the shikimic acid pathway for the biosynthesis of aromatic amino acids, chorismate mutase catalyzes the Claisen rearrangement of chorismate to prephenate with exquisite enantioselectivity and remarkable rate accelerations (up to  $k_{\text{cat}}/k_{\text{uncat}} = 10^6$ ).<sup>[94]</sup> Crystal structures of multiple organismic variants of this enzyme bound to transition-state analogues provide strong evidence that the amino acid side chains lining the active site engage chorismate with a network of attractive noncovalent interactions that cooperatively stabilize the transition state of the sigmatropic rearrangement (Figure 11 A).<sup>[94a,b]</sup> While hydrogen-bonding



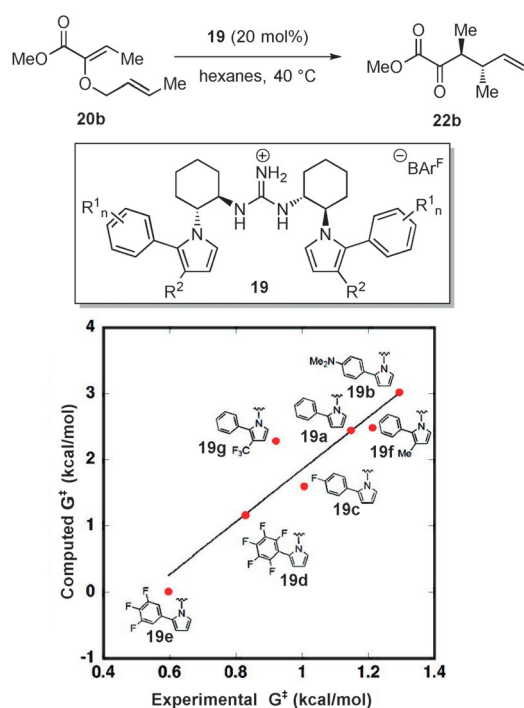
**Figure 11.** A) The BsCM active site binding to the Claisen rearrangement transition state as inferred from the crystal structure of the enzyme bound to a transition-state mimic. Calculated B) major and C) minor transition structures of the Claisen rearrangement catalyzed by guanidinium **22a**. Calculated with B3LYP/6-31G(d). Red dotted lines represent cation- $\pi$  interactions; blue dotted lines represent hydrogen bonds. Black or gray dotted lines indicate forming or breaking bonds. Key distances are shown in Å. C gray, H white, N blue, O red. Adapted with permission from Ref. [33c]. Copyright 2011 American Chemical Society.

interactions are dominant, evidence for an additional active-site cation- $\pi$  interaction has been advanced in the case of *Bacillus subtilis* chorismate mutase (BsCM).<sup>[95]</sup> Inspired by this enzymatic mechanism, Jacobsen and co-workers developed a series of  $C_2$ -symmetric guanidinium ion catalysts **19** for the Claisen rearrangements of  $\alpha$ - and  $\beta$ -ketoesters **20** and **21**, as discussed in Section 3.1.2.<sup>[33a,b]</sup> Crystallographic and computational analyses indicated that cation- $\pi$  interactions between the guanidinium  $\text{NH}_2$  hydrogen atoms and the pendent pyrrole rings control the ground-state catalyst conformation, framing the space around the hydrogen-bond donor with aromatic groups.<sup>[33c]</sup>

NMR titration experiments and computational studies further revealed that catalyst **19a** binds  $\alpha$ -ketoester **20** through dual hydrogen-bonding interactions to the ether and ester oxygen atoms. These stabilizing interactions are

strengthened in the polarized transition state as the partial negative charge on the enol fragment increases. Correspondingly, the allyl fragment accumulates significant partial positive charge, which can engage in a cation- $\pi$  interaction with the phenyl rings of the catalyst. DFT calculations indicated that this attractive cation- $\pi$  interaction exists in only one of the two diastereomeric transition states and is therefore responsible for enantiodifferentiation (Figure 11 B,C).

This finding subsequently informed the design of the more enantioselective catalyst **19b**, which bears electron-rich aryl groups capable of providing greater cation- $\pi$  stabilization in the enantiodetermining transition state. The experimentally observed relationship between catalyst substitution (catalysts **19a–19g**) and enantioselectivity was reproduced by DFT calculations, lending credence to the mechanistic model (Scheme 29). The cooperative action of the cationic hydro-



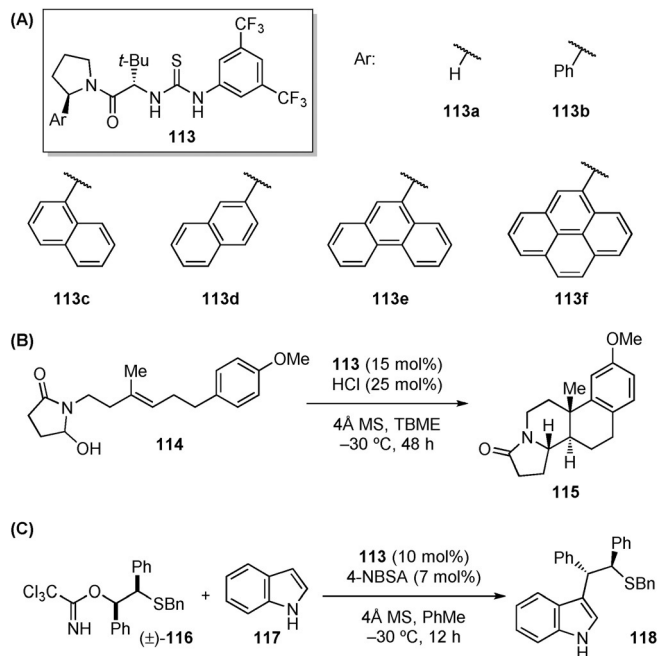
**Scheme 29.** Guanidinium-catalyzed asymmetric Claisen rearrangements. Relative free energies of activation were estimated according to classical transition-state theory ( $\Delta\Delta G^\ddagger = -RT \ln([(S,S)\text{-}22b]/[(R,R)\text{-}22b])$ ,  $T = 313.15\text{ K}$ ). Computed energies determined with M05-2X/6-31 g(d). The black line represents a least-squares fit to a linear function (intercept:  $-2.10$ ; slope:  $3.95$ ;  $R^2 = 0.88$ ). Adapted with permission from Ref. [33c]. Copyright 2011 American Chemical Society.

gen-bond donor and the electron-rich  $\pi$ -donor is thus directly analogous to the mechanism by which BsCM is proposed to catalyze the Claisen rearrangement in Nature.

#### 4.2.7. Reactions Catalyzed by Chiral Arylpyrrolidino Thioureas

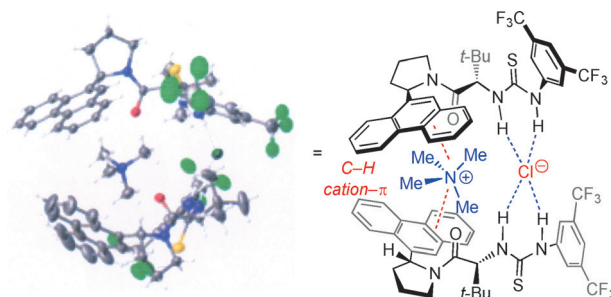
The guanidinium-catalyzed Claisen rearrangement described above operates by selective stabilization of a polarized transition structure bearing only partial charge separation. In

principle, the energetic benefits of catalysis utilizing cooperative cation- $\pi$  and hydrogen-bonding interactions can be amplified in reactions involving fully charged ion-pair intermediates. Motivated by this postulate, Jacobsen and co-workers developed a family of thiourea hydrogen-bond donor catalysts (**113**) bearing precisely positioned aromatic substituents (Scheme 30 A).<sup>[96]</sup> The crystal structures of these



**Scheme 30.** A) Chiral arylpyrrolidine thiourea catalysts for enantioselective B) polycyclization and C) episulfonium ring-opening reactions. 4-NBSA = 4-nitrobenzenesulfonic acid, TBME = *tert*-butyl methyl ether.

catalysts bound to tetramethylammonium chloride provide compelling evidence for specific attractive interactions with both components of the ion pair (Figure 12).<sup>[96b]</sup> The thiourea binds the chloride anion through hydrogen-bonding interactions while the ammonium cation associates with the  $\pi$ -face of the arylpyrrolidine. As outlined below, this type of ion-pair recognition has also been extended successfully to the



**Figure 12.** Crystal structure of thiourea **113e** binding to tetramethylammonium chloride with a 2:1 stoichiometry. Red dotted lines represent cation- $\pi$  interactions; blue dotted lines represent hydrogen bonds. C gray, H white, N blue, O red, F green, S yellow.

selective stabilization of transition states, thereby enabling enantioselective reactions.

#### 4.2.7.1. Chiral Thiourea Catalyzed Cationic Polycyclization Reactions

The utility of these arylpyrrolidine thiourea catalysts was first demonstrated in the alkylation of oxocarbenium ions<sup>[97]</sup> and was, soon after, reinforced with catalysis of the cationic polycyclization of hydroxylactam derivatives.<sup>[98]</sup> Catalyst structure–enantioselectivity relationship studies revealed a linear free energy relationship (LFER) between the polarizability of the aryl substituent of the catalyst and the enantioselectivity of the cyclization of **114** (Scheme 30B and Table 6). Furthermore, analysis of temperature effects on

**Table 6:** Dependence of the reactivity and enantioselectivity on arene expanse in the arylpyrrolidine thiourea catalyzed polycyclization and episulfonium ring-opening reactions.<sup>[a]</sup>

| Entry | Product    | Yield [%] (ee [%]) with |             |             |             |             |             |
|-------|------------|-------------------------|-------------|-------------|-------------|-------------|-------------|
|       |            | <b>113a</b>             | <b>113b</b> | <b>113c</b> | <b>113d</b> | <b>113e</b> | <b>113f</b> |
| 1     | <b>115</b> | n.d.                    | 12 (25)     | 46 (60)     | 33 (61)     | 52 (87)     | 78 (95)     |
| 2     | <b>118</b> | 15 (12)                 | 72 (73)     | 84 (84)     | 80 (85)     | 93 (93)     | 91 (91)     |

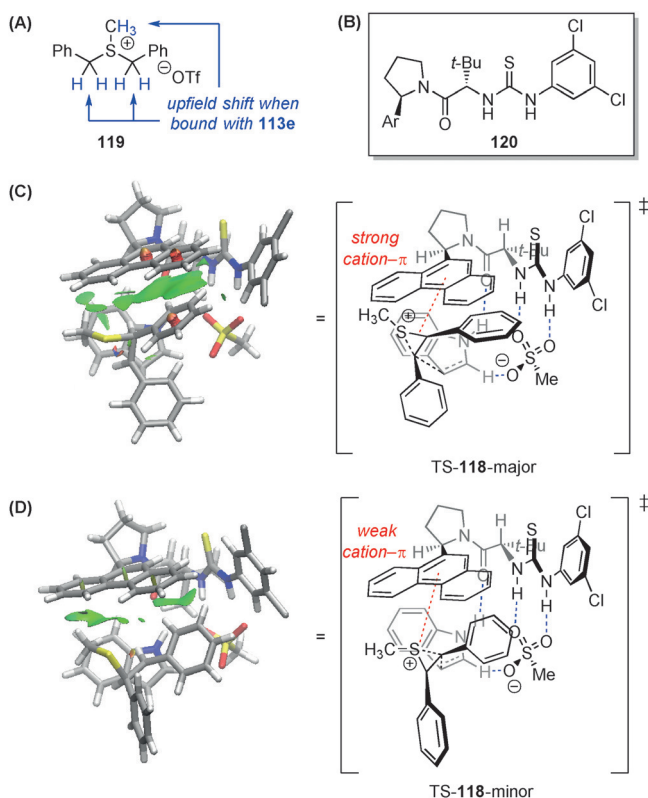
[a] See Scheme 30 for the catalyst and product structures.

enantioselectivity revealed that the improved stereinduction afforded by catalysts bearing expanded aromatic substituents is due to the increased differential enthalpic stabilization of the major transition structure. Taken together, these results provide compelling evidence that attractive cation– $\pi$  interactions between the cyclization transition structure and the catalyst serve to dictate the stereochemical outcome of the cyclization cascade. This mechanism of catalysis is loosely analogous to that employed by oxidosqualene cyclase, which stabilizes the highly delocalized cationic transition structure for oxidosqualene polycyclization through multiple interactions with aromatic residues lining the active site.<sup>[13]</sup>

#### 4.2.7.2. Chiral Thiourea Catalyzed Episulfonium Ring-Opening Reactions

Lin and Jacobsen subsequently extended this strategy for cooperative noncovalent catalysis to the desymmetrization of *meso*-episulfonium ions (formed in situ from **116**) with indole derivatives (**117**; Scheme 30C).<sup>[99]</sup> As for the cationic polycyclization reactions, enantioselectivity is highly dependent on the expanse of the  $\pi$ -system of the catalyst, with the 9-phenanthryl-substituted catalyst **113e** affording product **118** with the highest level of enantioinduction (Table 6). Furthermore, comparison of the absolute rates of reaction with different catalysts revealed that enantioselectivity correlates directly with rate acceleration. As C–C bond formation is both rate- and enantiodetermining across the catalyst series (**113a**–**113f**), this observation unambiguously demonstrates that the arene-dependent increase in enantioselectivity is due to the selective stabilization of the major transition state rather than to the destabilization of the minor transition state.

NMR spectroscopic analysis of thiourea catalysts **113e** and **113a** in the presence of episulfonium model compound **119** provided insight into the nature of the stabilizing interactions (Figure 13). The resonances of the benzylic and methyl protons of **119** undergo a significant upfield shift upon complexation with **113e**, whereas these protons are not measurably perturbed upon complexation with **113a**, which lacks an aryl substituent. The relevance of these ground-state interactions in the enantiodetermining transition state was supported by computational (DFT) studies of simplified catalysts (**120b**, **120c**, **120e**, and **120f**).<sup>[100]</sup>



**Figure 13.** A) The benzylic proton  $^1\text{H}$  NMR resonances of episulfonium model **119** undergo a significant upfield shift upon complexation with **113e**. B) Simplified catalysts used to approximate **113** in computational analyses. The lowest-energy computed transition structures (M05-2X/6-31G(d)) with catalyst **120e** en route to the C) major and D) minor enantiomers of product **118**. Noncovalent interactions (green surfaces) were visualized with NCIPLOT. Red dotted lines represent cation– $\pi$  interactions; blue dotted lines represent hydrogen bonds. Black or gray dotted lines indicate forming or breaking bonds. C gray, H white, N blue, O red, S yellow. Adapted from Ref. [100].

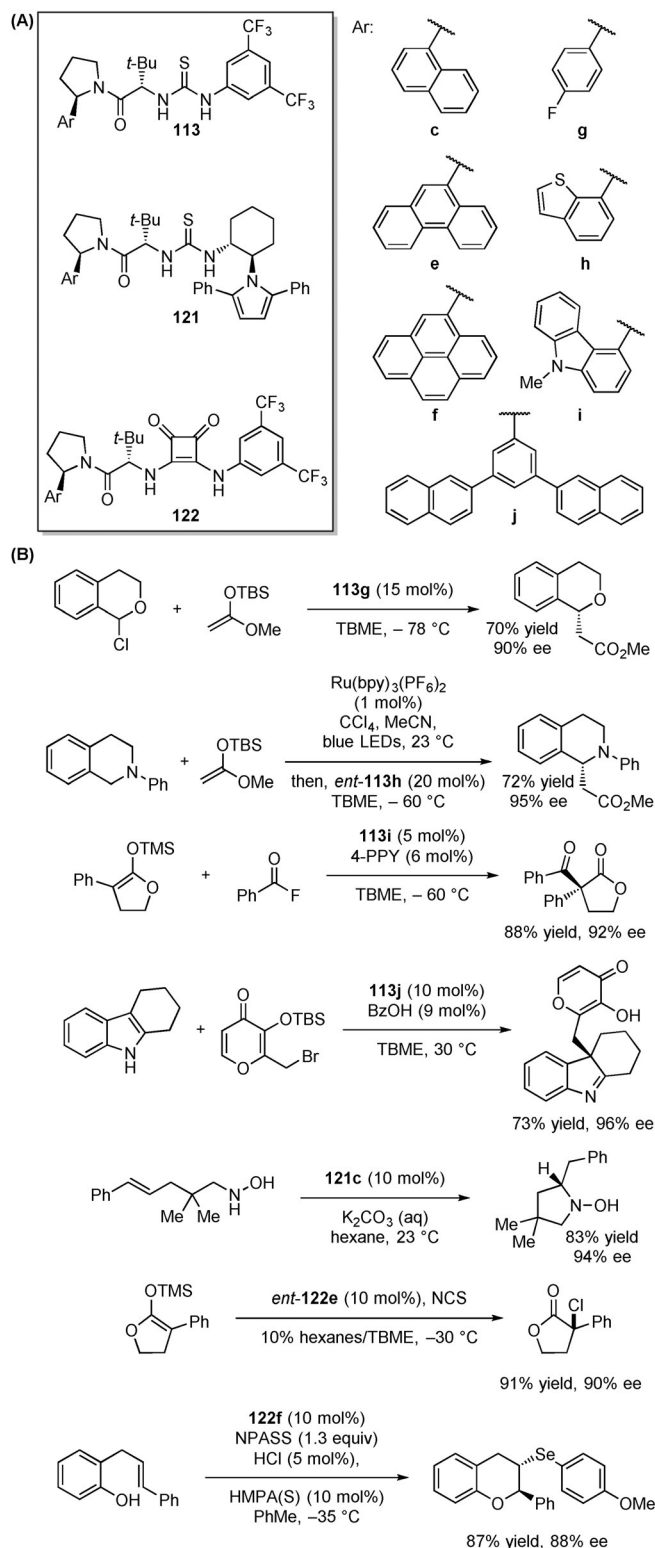
The lowest-energy diastereomeric transition structures calculated for the ring opening in the presence of catalyst **120e** differ primarily in terms of the distance between the aromatic group of the catalyst and the cationic reactant. Noncovalent interaction analysis<sup>[101]</sup> of these structures indicates that an attractive interaction exists between the  $\pi$ -face of the phenanthryl group and the reacting episulfonium ion in the major transition structure but is almost completely absent

in the minor transition structure. Taken together, these experimental and computational studies are fully consistent with the conclusion that the differential strength of the cation– $\pi$  interaction between the catalyst and episulfonium ion in the major and minor transition states is responsible for enantioinduction. The mechanism of the thiourea-catalyzed alkylation of indoles with episulfonium electrophiles is highly reminiscent of alkylations mediated by SAM-dependent methyltransferases as well as Dougherty's cyclophanes (Section 4.2.1), which rely on similar cation– $\pi$  interactions to engage the cofactor in the ground and transition state.

Since the initial disclosure of catalysts with the general structure depicted in Scheme 30 (**113b–113f**) this family of arylpyrrolidine-derived dual hydrogen-bond donor catalysts has been applied by Jacobsen and co-workers to a range of highly enantioselective transformations aside from those outlined above. In these studies, the hydrogen-bond donor is generally hypothesized to activate a neutral substrate by binding and/or abstracting an achiral, anionic leaving group (typically a halide, carboxylate, or sulfonate) to generate a reactive cationic electrophile. The rest of the catalyst structure, and the aryl group of the pyrrolidino amide in particular, is proposed to provide stabilizing secondary interactions in selectivity-determining events, leading to enantioselective nucleophilic addition. Examples include the alkylation of isoquinolines,<sup>[102]</sup> the C acylation of silyl ketene acetals,<sup>[103]</sup> the alkylation of pyrone derivatives,<sup>[104]</sup> Cope-type hydroaminations of olefins,<sup>[105]</sup> the electrophilic chlorination of silyl ketene acetals,<sup>[106]</sup> and selenoetherification reactions of olefins (Scheme 31).<sup>[107]</sup> In each of these cases, the identity of the arylpyrrolidine substituent exerts a strong, if not defining, influence on the enantioselectivity. On a broader level, these examples illustrate how lessons from enzymology and molecular recognition can inform and inspire applications of noncovalent interactions, such as the cation– $\pi$  interaction, for the development of productive strategies in small-molecule catalysis.

## 5. Summary and Outlook

Since its discovery in the 1980s, the importance of the cation– $\pi$  interaction has become increasingly recognized in many fields of chemical science. Recent efforts, grounded in the extensive characterization of the energetics of cation– $\pi$  interactions, have sought to incorporate these attractive non-covalent interactions into functional catalyst systems in order to influence the rate and selectivity of chemical reactions through the binding of reaction intermediates and the differential stabilization of transition structures. This Review has highlighted examples of small-molecule catalyst systems in which cation– $\pi$  interactions have been invoked in the mechanism of catalysis, granting emphasis to cases in which the nature and role of the cation– $\pi$  interaction have been evaluated in detail. Experimental techniques used for the elucidation of these effects, including kinetic analysis, modern spectroscopic methods, and crystallographic structural analysis in combination with state-of-the-art computational methods, have been critical in this endeavor. Indeed, the



**Scheme 31.** Representative reactions catalyzed by arylpyrrolidine-derived hydrogen-bond donor catalysts. HMPA-(S) = tris(dimethylamino)phosphorus sulfide, NPASS = *N*-para-anisylsuccinimide, 4-PPY = 4-pyrrolidinopyridine.

interplay of theory and experiment is likely to continue to play a crucial role in the exploration of applications of the cation– $\pi$  interaction in small-molecule catalysis. Research in

this field is just beginning to flourish, and its scope will undoubtedly continue to expand with the identification of new modes of reactivity, the design of novel catalyst structures, and the refinement of our current understanding of these noncovalent interactions.

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